Name:____________________________________ Period:_____

Assignment: ESS Skills Bootcamp

CASE FILE 1: The Retreat of Mendenhall Glacier

Essential Question: How do scientists know what happened if they weren't there to see it?

Background: Located near Juneau, Alaska, Mendenhall Glacier has been carefully monitored by scientists for decades. Historical photographs, satellite images, and field measurements all suggest that the glacier has changed significantly over time.

Task: Your investigation team has been asked to review the available evidence to determine whether the glacier is retreating. Before scientists can draw conclusions, they must separate evidence from assumptions and determine whether the available information supports the claim.

Directions: Read the case file, examine each exhibit carefully, and answer the questions using only the information provided. Support your answers with evidence whenever possible.

Before You Investigate

- **Evidence:** information collected through **observations, measurements, investigations, or data**. Scientists use evidence to support or challenge a scientific claim.
- **Assumptions: ideas** accepted as true based on current scientific knowledge. They help scientists interpret evidence and build explanations, but assumptions should always be supported by evidence whenever possible.

Remember: Scientific conclusions should be supported by evidence, not assumptions.

Student Investigation Step 1: *Analyze the Evidence*

1. Compare Exhibits A and B. Identify two pieces of evidence that suggest the glacier has changed over time.

2. Which claim is best supported by the evidence?
   A. The glacier has grown larger over time.
   B. The glacier has remained about the same size.
   C. The glacier has decreased in size over time.
   D. There is not enough evidence to determine if the glacier has changed.

Student Investigation Step 2: *Evaluate the Investigation*

3. Decide whether each statement is Evidence (E) or an Assumption (A).

____a. The glacier is shorter today than it was 40 years ago.
____b. The photographs accurately represent the glacier.
____c. Average annual temperatures have increased.
____d. The measuring equipment was working correctly.
____e. Ice thickness measurements show a decrease over time.
____f. The glacier has remained in the same location throughout the study.

Student Investigation Step 3: *Draw a Conclusion*

4. Which two exhibits provide the strongest evidence that Mendenhall Glacier has changed over time? Explain your answer.

5. If you were the lead scientist, what additional evidence would you collect? Explain why.

Student Investigation Step 4: *Think Like a Scientist*

6. Why is it important for scientists to distinguish between evidence and assumptions before reaching a conclusion?

☐ **Analyzed Evidence**

Name:____________________________________ Period:_____ **Assignment: ESS Skills Bootcamp**

CASE FILE 2: Lake Effect on the Way

Essential Question: How do scientists use graphs and data tables to identify patterns and draw conclusions?

Background: Meteorologists with the National Weather Service (NWS) collect data from weather stations, satellites, radar, and other instruments to monitor changing weather conditions. Instead of relying on a single measurement, they compare multiple sources of data to identify patterns and better understand what is happening in the atmosphere. Your investigation team has been asked to examine recent weather data collected in western New York and determine what the information reveals about an approaching weather system.

Task: Analyze the exhibits to identify patterns, interpret the data, and draw conclusions about the changing weather conditions.

Directions: Read the case file, examine each exhibit carefully, and answer the questions using only the information provided. Support your answers with evidence whenever possible.

Before You Investigate

- **Data:** facts, measurements, or observations collected during an investigation.
- **Pattern:** a repeated trend or relationship that can help scientists explain events or make predictions.
- **Trend:** the general direction that data change over time.

Remember: Scientists examine multiple sources of data before drawing conclusions.

Student Investigation Step 1: ***Analyze the Evidence***

1. Identify two patterns in the data and explain what each pattern suggests about the changing weather conditions.

2. Which exhibit provides the strongest evidence that weather conditions are changing? Explain your reasoning using the exhibits.

Student Investigation Step 2: ***Evaluate the Investigation***

3. Which exhibit would be most useful for predicting tomorrow's weather? Explain why another exhibit alone would not be enough.

4. Why is it important for scientists to examine both graphs and data tables instead of relying on only one source of information?

Student Investigation Step 3: ***Draw a Conclusion***

5. Based on the evidence, what conclusion can you make about the changing weather conditions?
Support your answer using at least two exhibits.

Student Investigation Step 4: ***Think Like a Scientist***

6. How does organizing information into graphs and tables help scientists better understand natural phenomena?

7. A meteorologist claims that a storm will definitely arrive tomorrow because temperatures dropped today. Is this claim fully supported by the available evidence? Explain why or why not using the exhibits.

☐ **Analyzed Data**

Name:________________________________ Period:_____

Assignment: ESS Skills Bootcamp

CASE FILE 3: Mapping the San Andreas Fault

Essential Question: How do scientists use models to study things they cannot observe directly?

Background: The San Andreas Fault is one of the most well-known faults in the world. Scientists from the U.S. Geological Survey (USGS) study the fault to better understand how Earth's crust moves and where future earthquakes may occur. Because scientists cannot directly observe deep beneath Earth's surface, they rely on scientific models to represent underground features and explain how Earth's systems work.

Task: Your investigation team has been asked to examine several exhibits related to the San Andreas Fault. Determine which exhibits are scientific models, what information those models provide, and how they help scientists better understand Earth's crust.

Directions: Read the case file, examine each exhibit carefully, and answer the questions using only the information provided. Support your answers with evidence whenever possible.

Before You Investigate

- **Model:** a simplified **representation** of an object, system, or process that helps scientists describe, explain, or predict how something works.
- Every model has **limitations**. A model cannot include every detail of the real object or system it represents.

Remember: Scientists often use multiple models because each model provides different information.

Student Investigation Step 1: ***Analyze the Evidence***

1. Which exhibits are scientific models? Explain how you know.

2. Which exhibit would be most useful for understanding what is happening beneath Earth's surface? Explain your reasoning.

Student Investigation Step 2: ***Evaluate the Investigation***

3. Why would scientists use more than one model to study the San Andreas Fault?
   A. One model can show every detail of the fault.
   B. Different models provide different types of information.
   C. Scientists are required to use multiple models.
   D. Models are more accurate than real observations.

4. Which statement best describes a limitation of scientific models?
   A. Models always include every detail of the real system.
   B. Models cannot be updated with new evidence.
   C. Models simplify reality and may leave out important details.
   D. Models can only be used in Earth science.

Student Investigation Step 3: ***Draw a Conclusion***

5. Explain how using both observations and models helps scientists better understand the San Andreas Fault. Use evidence from the exhibits to support your answer.

Student Investigation Step 4: ***Think Like a Scientist***

6. Imagine scientists discover new evidence that changes their understanding of the San Andreas Fault. How might this affect the models they use?

☐ **Used Models**

Name:______________________________________ Period:_____

Assignment: ESS Skills Bootcamp

CASE FILE 4: Why Are These Plants Dying?

Essential Question: How do scientists design fair investigations to answer scientific questions?

Background: Researchers at the Buffalo and Erie County Botanical Gardens noticed that one group of plants was growing much more slowly than the others. The scientists wanted to determine which environmental factor was affecting plant growth. To answer this question, they designed an investigation. Your investigation team has been asked to evaluate the investigation and determine whether it was designed fairly.

Task: Review the investigation and identify any problems with the experimental design. Determine how the investigation could be improved.

Directions: Read the case file, examine each exhibit carefully, and answer the questions using only the information provided. Support your answers with evidence whenever possible.

Before You Investigate

- **Independent Variable:** the factor that is intentionally changed by the scientist.
- **Dependent Variable:** the factor that is measured or observed during the investigation.
- **Controlled Variables:** the factors that are kept the same so the investigation is fair.

Remember: A fair investigation changes only one variable at a time.

Student Investigation Step 1: ***Analyze the Evidence***

1. What question are the scientists trying to answer?

2. Identify the:

- Independent Variable:_________________________
- Dependent Variable:_________________________
- Two Controlled Variables:_________________________&_________________________

Student Investigation Step 2: ***Evaluate the Investigation***

3. One group of plants received more sunlight and more water than the other groups. Why is this a problem?

A. The investigation lasted too long.
B. More than one variable was changed.
C. The plants were measured too often.
D. There were too many plants.

4. Describe one improvement the scientists could make to strengthen the investigation.

Student Investigation Step 3: ***Draw a Conclusion***

5. Based on the investigation, can the scientists confidently determine what caused the difference in plant growth? Explain your reasoning using evidence.

Student Investigation Step 4: ***Think Like a Scientist***

6. Why is controlling variables important in scientific investigations?

☐Designed Investigations

Name:_________________________________ Period:_____ **Assignment: ESS Skills Bootcamp**

CASE FILE 5: The Mystery of the Green Water

Essential Question: How do scientists use systems thinking to explain changes in Earth's environment?

Background: During late summer, parts of Lake Erie sometimes turn bright green due to harmful algal blooms. Scientists from the Great Lakes Environmental Research Laboratory (GLERL) study the lake to understand why these blooms develop and how they affect the ecosystem. Rather than studying one factor at a time, scientists examine the entire system and how its parts interact.

Task: Your investigation team has been asked to examine the Lake Erie system. Use the exhibits to identify the components of the system, explain how they interact, and determine how changes in one part of the system affect the others.

Directions: Read the case file, examine each exhibit carefully, and answer the questions using only the information provided. Support your answers with evidence whenever possible.

Before You Investigate
- **System:** a group of interacting parts that work together.
- **Interaction:** occurs when one part of a system affects another.
- **Input:** something that enters a system.
- **Output:** something that leaves a system.

Re*member: A change in one part of a system can affect the entire system*

Student Investigation Step 1: ***Analyze the Evidence***

1. Identify four components of the Lake Erie system shown in the exhibits.

2. Describe one interaction occurring within the system.

Student Investigation Step 2: ***Evaluate the Investigation***

3. Which statement best describes a system?
   A. A collection of unrelated objects.
   B. A group of interacting parts that work together.
   C. A single event that happens once.
   D. A list of observations.

4. Which exhibit best demonstrates how one part of the system affects another? Explain your reasoning.

Student Investigation Step 3: ***Draw a Conclusion***

5. Explain how a change in one component of the Lake Erie system could affect another component. Use evidence from the exhibits.

Student Investigation Step 4: ***Think Like a Scientist***

6. Why is it important for scientists to study entire systems instead of only one part?

☐**Applied Systems Thinking**

Name:____________________________________ Period:_____

Assignment: ESS Skills Bootcamp

CASE FILE 6: Has NASA Found a New Planet?

Essential Question: How do scientists decide whether evidence supports a scientific claim?

Background: NASA's Transiting Exoplanet Survey Satellite (TESS) searches for planets orbiting distant stars. Scientists cannot see most of these planets directly. Instead, they analyze data collected by telescopes and determine whether the evidence supports the claim that a planet exists. Sometimes, different scientists interpret the same evidence in different ways. Before announcing a discovery, they must determine whether the available evidence is strong enough to support their claim.

Task: Review the exhibits and determine whether the available evidence supports the claim that a new planet has been discovered.

Directions: Read the case file, examine each exhibit carefully, and answer the questions using only the information provided. Support your answers with evidence whenever possible.

Before You Investigate

- **Claim:** a statement or conclusion that answers a scientific question.
- **Evidence:** includes observations, measurements, investigations, or data used to support or challenge a claim.
- **Reasoning:** explains why the evidence supports the claim using scientific thinking.

Remember: A strong scientific explanation includes a clear claim supported by relevant evidence and logical reasoning.

Student Investigation Step 1: ***Analyze the Evidence***

1. What pattern do you observe in the brightness graph?

2. Which exhibit provides the strongest evidence for the claim that a planet may exist? Explain your reasoning.

Student Investigation Step 2: ***Evaluate the Investigation***

3. One scientist concludes, "A new planet has definitely been discovered." Another scientist says, "The evidence is promising, but more data are needed." Which conclusion is better supported by the evidence? Explain your reasoning.

4. Which additional piece of evidence would strengthen the claim that a planet exists?
   A. More observations showing the same pattern over time
   B. A photograph of another star
   C. The distance to the Moon
   D. Today's weather conditions

Student Investigation Step 3: ***Draw a Conclusion***

5. Write a scientific explanation using:
   - Claim:
   - Evidence:
   - Reasoning:

Student Investigation Step 4: ***Think Like a Scientist***

6. Why should scientists continue collecting evidence even after they think they have found an answer?

☐ **Constructed Explanations**

Name:______________________________________ Period:_____

Assignment: ESS Skills Bootcamp

CASE FILE 7: Mapping the Lava Flow

Essential Question: How do scientists use maps to interpret and communicate information?

Background: Scientists from the U.S. Geological Survey (USGS) are monitoring Kīlauea Volcano in Hawaii. During recent eruptions, lava flowed across the landscape, threatening roads and nearby communities. Maps allow scientists to identify locations, recognize patterns, communicate hazards, and make informed decisions during emergencies.

Task: Use the exhibits to determine where the lava has traveled, identify areas at risk, and explain how maps help scientists communicate important information.

Directions: Read the case file, examine each exhibit carefully, and answer the questions using only the information provided. Support your answers with evidence whenever possible.

Before You Investigate

- **Map:** a representation of an area that shows the locations of natural or human-made features.
- **Map scale:** shows the relationship between distances on a map and actual distances on Earth's surface.
- **Legend:** explains the symbols and colors used on a map.

Remember: Scientists use maps to identify locations, recognize patterns, and communicate information.

Student Investigation Step 1: ***Analyze the Evidence***

1. Which direction did the lava flow move over time? Use evidence from the exhibits to support your answer.

2. Which exhibit would be most useful for estimating the real-world distance the lava traveled? Explain your reasoning.

Student Investigation Step 2: ***Evaluate the Investigation***

3. Why is a map legend important?
  A. It measures elevation.
  B. It explains the meaning of symbols and colors.
  C. It predicts future eruptions.
  D. It measures temperature.

4. Explain why scientists would use both a map and a satellite image when monitoring a volcanic eruption.

Student Investigation Step 3: ***Draw a Conclusion***

5. Based on the exhibits, which area appears to be at the greatest risk from future lava flows? Support your answer with evidence.

Student Investigation Step 4: ***Think Like a Scientist***

6. How do maps help scientists make decisions during natural hazards?

☐ **Interpreted Maps**

Name:____________________________________ Period:_____

Assignment: ESS Skills Bootcamp

CASE FILE 8: Following the Wildfire

Essential Question: How do scientists use multiple sources of evidence to build reliable conclusions?

Background: Scientists studying the 2023 Canadian wildfires used many different sources of information to monitor the fires and predict how smoke would affect nearby communities, including parts of New York State.
No single source of information provided the complete picture. Instead, scientists combined observations from satellites, maps, weather data, and air quality measurements to better understand the situation.

Task: Analyze the exhibits to determine how each source of information contributes to the investigation. Decide which exhibits provide the strongest evidence and explain why scientists use multiple sources when studying natural events.

Directions: Read the case file, examine each exhibit carefully, and answer the questions using only the information provided. Support your answers with evidence whenever possible.

Before You Investigate

- **Multiple Sources of Evidence:** Scientists collect information from many different sources, including observations, measurements, maps, graphs, photographs, satellite images, and models.
- **Corroborate:** to compare multiple sources of information to determine whether they support the same conclusion.

Remember: Scientists are more confident in a conclusion when multiple sources of evidence support it.

Student Investigation Step 1: ***Analyze the Evidence***

1. Which two exhibits provide the strongest evidence that wildfire smoke affected western New York?
Explain your reasoning.

2. What information does one exhibit provide that another does not?

Student Investigation Step 2: ***Evaluate the Investigation***

3. A scientist only looks at the satellite image before drawing a conclusion. Why is this not the best scientific approach?
A. Satellite images are never accurate.
B. One source of information rarely tells the complete story.
C. Scientists should never use satellite images.
D. Maps are always more accurate than photographs.

4. Which additional source of information would strengthen the investigation the most?
Explain your answer.

Student Investigation Step 3: ***Draw a Conclusion***

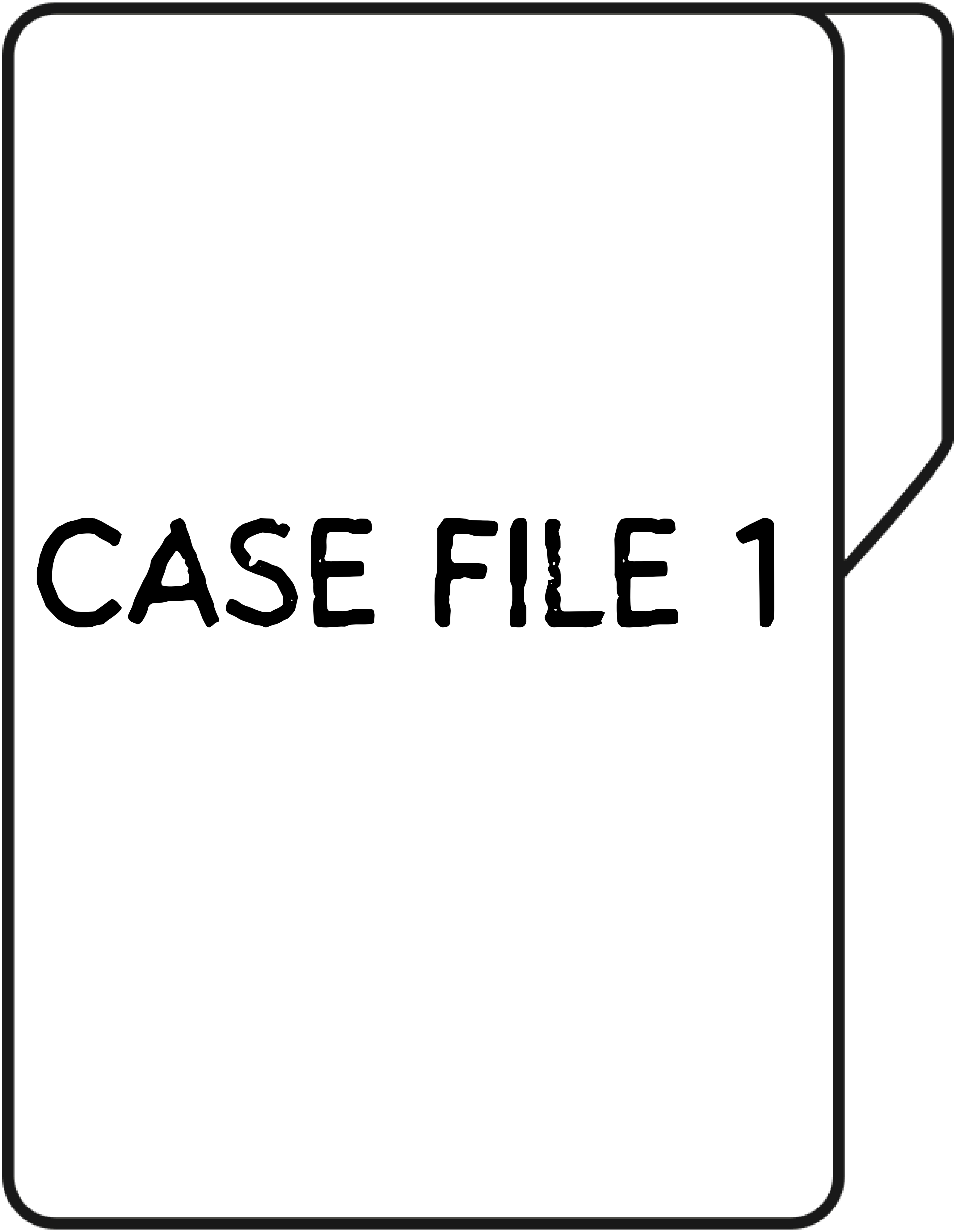
5. Explain why scientists use multiple sources of information when investigating natural events.
Support your answer using evidence from the exhibits.

Student Investigation Step 4: ***Think Like a Scientist***

6. Which exhibit did you find most useful? Explain why it was important to your investigation.

☐ **Evaluated Scientific Information**

CASE FILE 1



CASE FILE 1

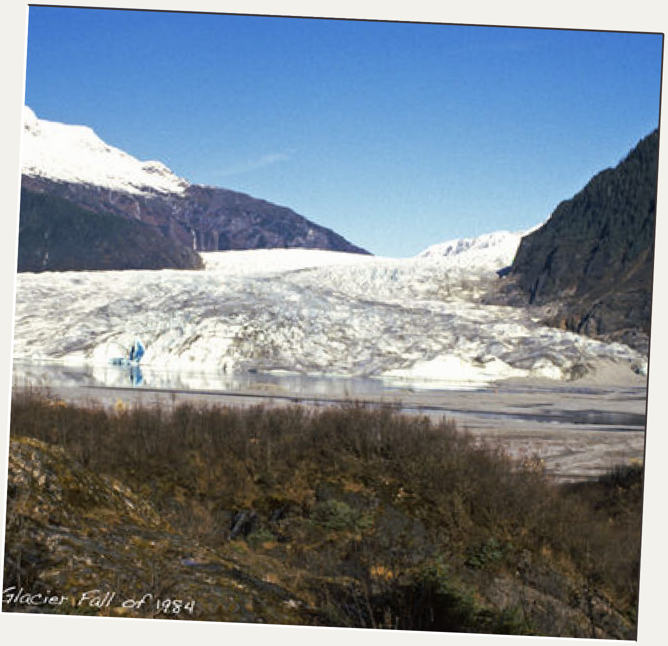

Glacier Fall of 1984

Exhibit A
Photograph of Mendenhall Glacier (1984)

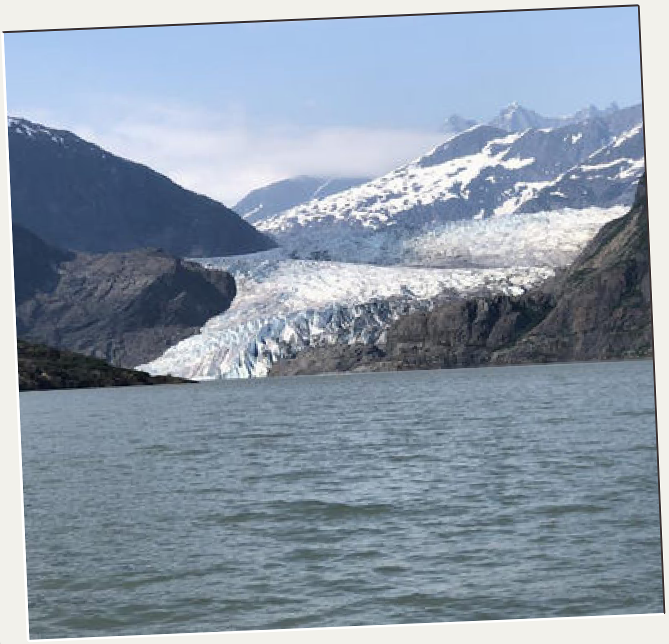


Exhibit B
Photograph of Mendenhall Glacier (2024)

CASE FILE 1

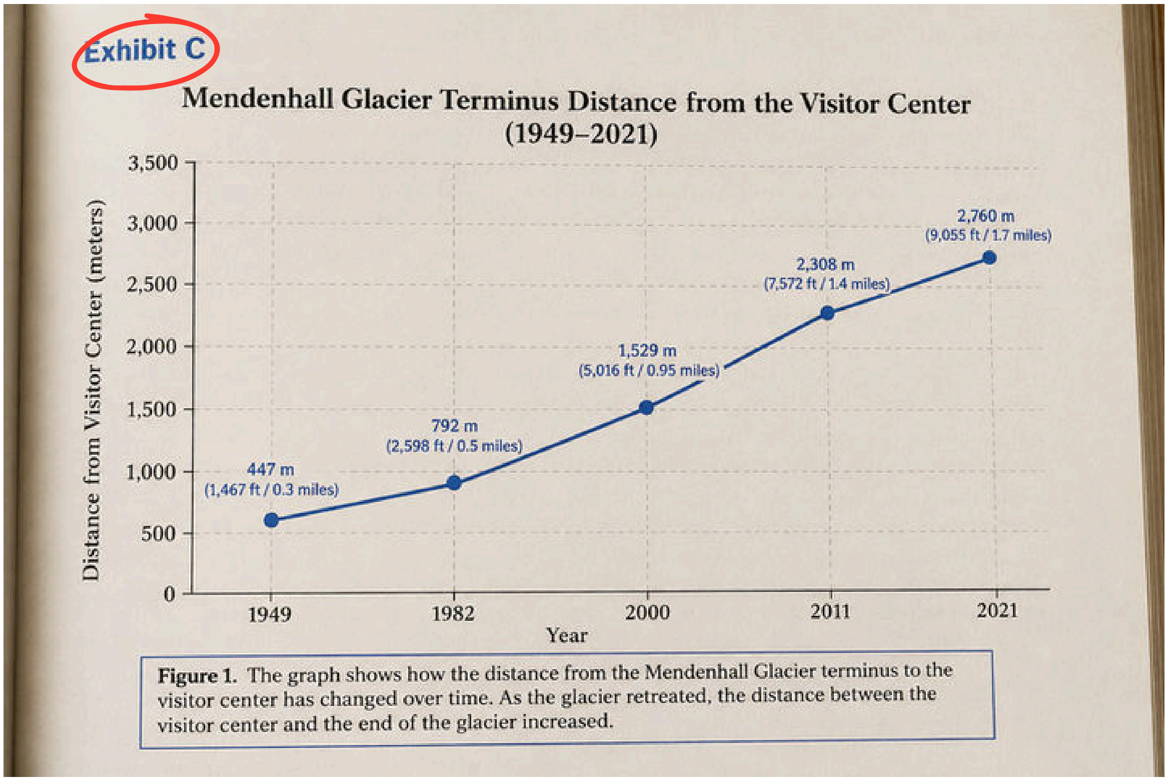


Exhibit C

Mendenhall Glacier Terminus Distance from the Visitor Center (1949–2021)

Figure 1. The graph shows how the distance from the Mendenhall Glacier terminus to the visitor center has changed over time. As the glacier retreated, the distance between the visitor center and the end of the glacier increased.

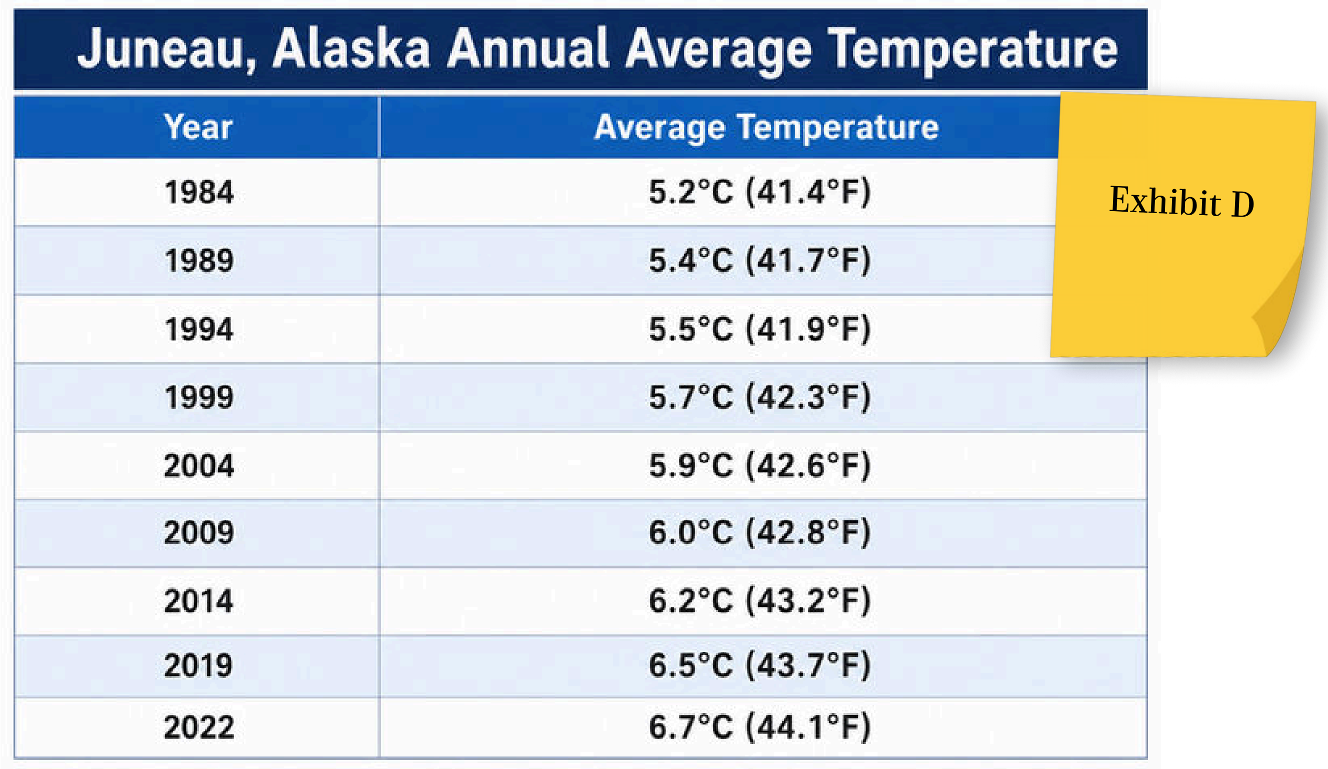
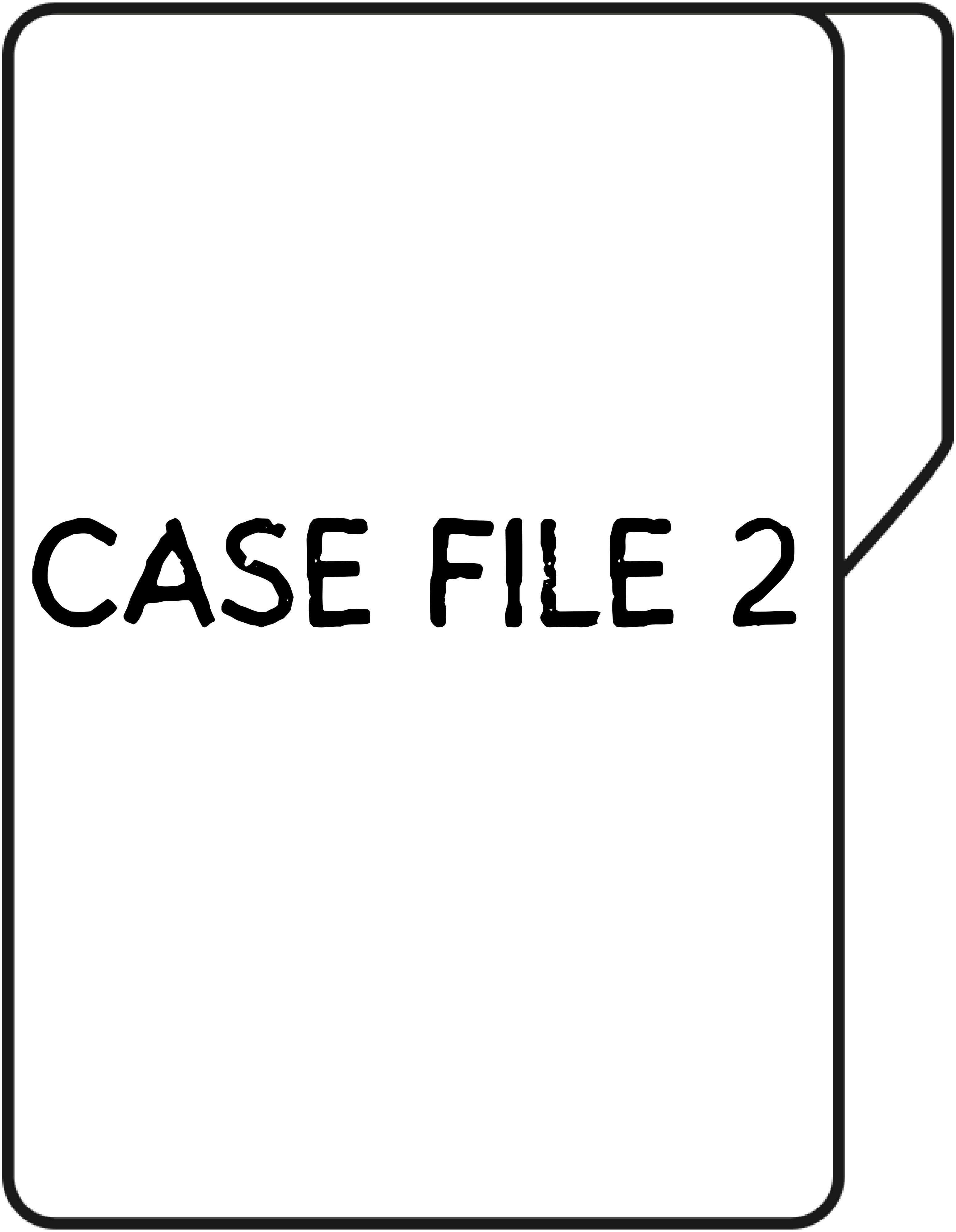


Exhibit D

Juneau, Alaska Annual Average Temperature

Year	Average Temperature
1984	5.2°C (41.4°F)
1989	5.4°C (41.7°F)
1994	5.5°C (41.9°F)
1999	5.7°C (42.3°F)
2004	5.9°C (42.6°F)
2009	6.0°C (42.8°F)
2014	6.2°C (43.2°F)
2019	6.5°C (43.7°F)
2022	6.7°C (44.1°F)



CASE FILE 2

CASE FILE 2

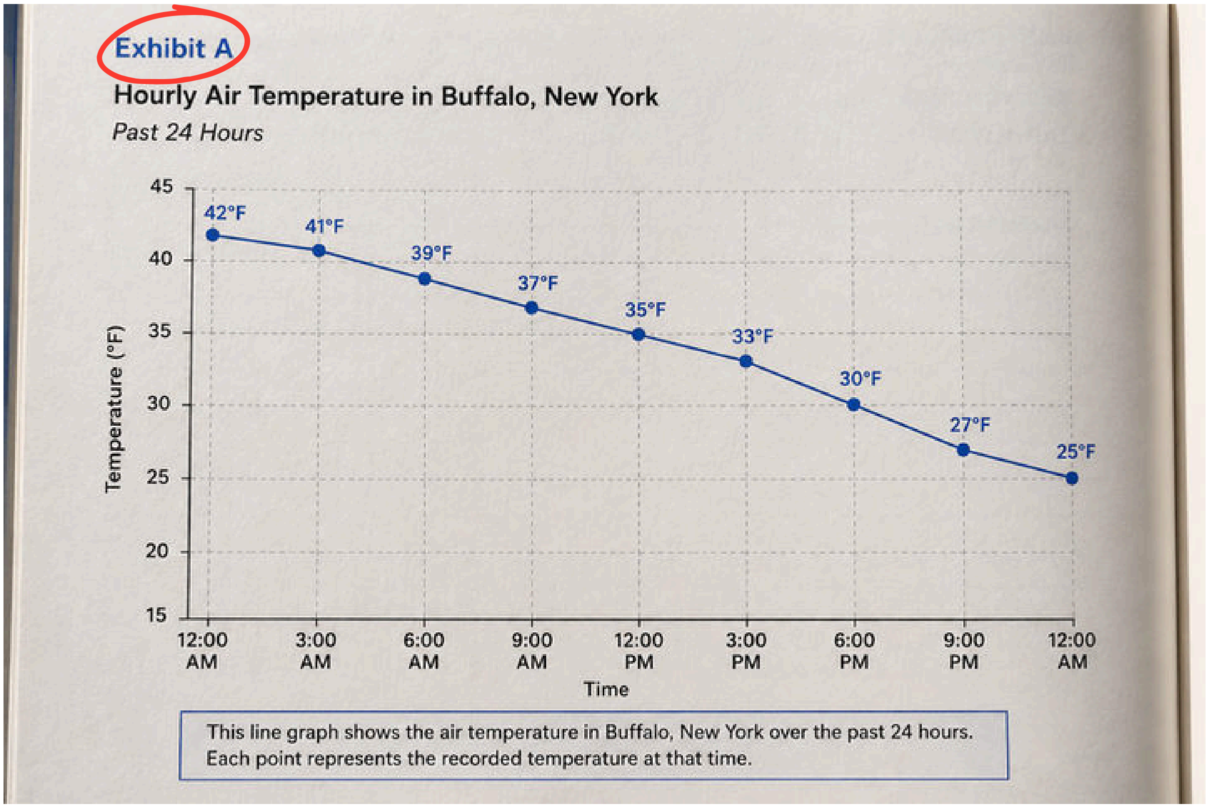


Exhibit A

Hourly Air Temperature in Buffalo, New York

Past 24 Hours

This line graph shows the air temperature in Buffalo, New York over the past 24 hours. Each point represents the recorded temperature at that time.

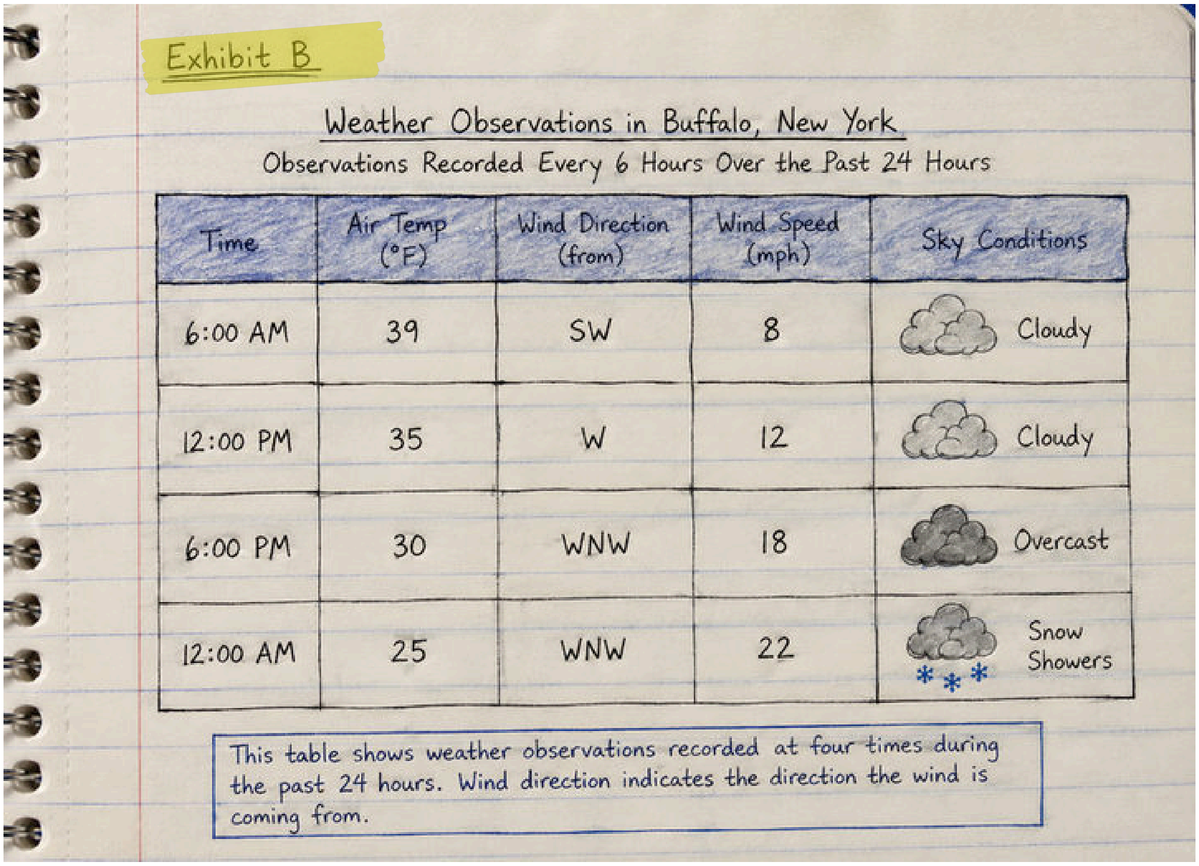


Exhibit B

Weather Observations in Buffalo, New York

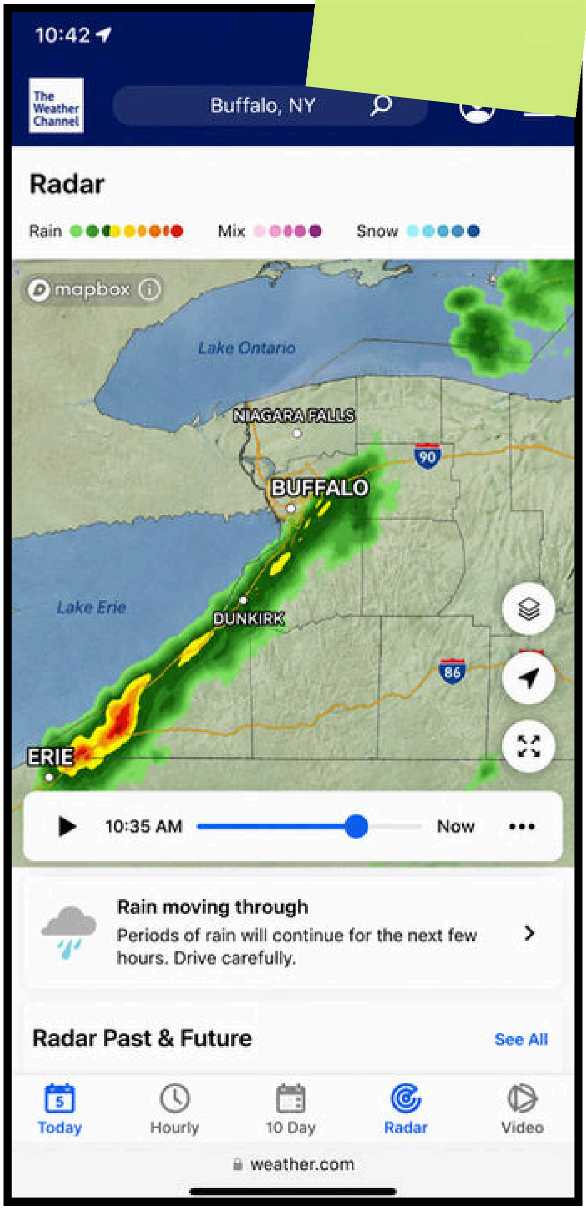
Observations Recorded Every 6 Hours Over the Past 24 Hours

Time	Air Temp (°F)	Wind Direction (from)	Wind Speed (mph)	Sky Conditions
6:00 AM	39	SW	8	Cloudy
12:00 PM	35	W	12	Cloudy
6:00 PM	30	WNW	18	Overcast
12:00 AM	25	WNW	22	Snow Showers

This table shows weather observations recorded at four times during the past 24 hours. Wind direction indicates the direction the wind is coming from.

CASE FILE 2

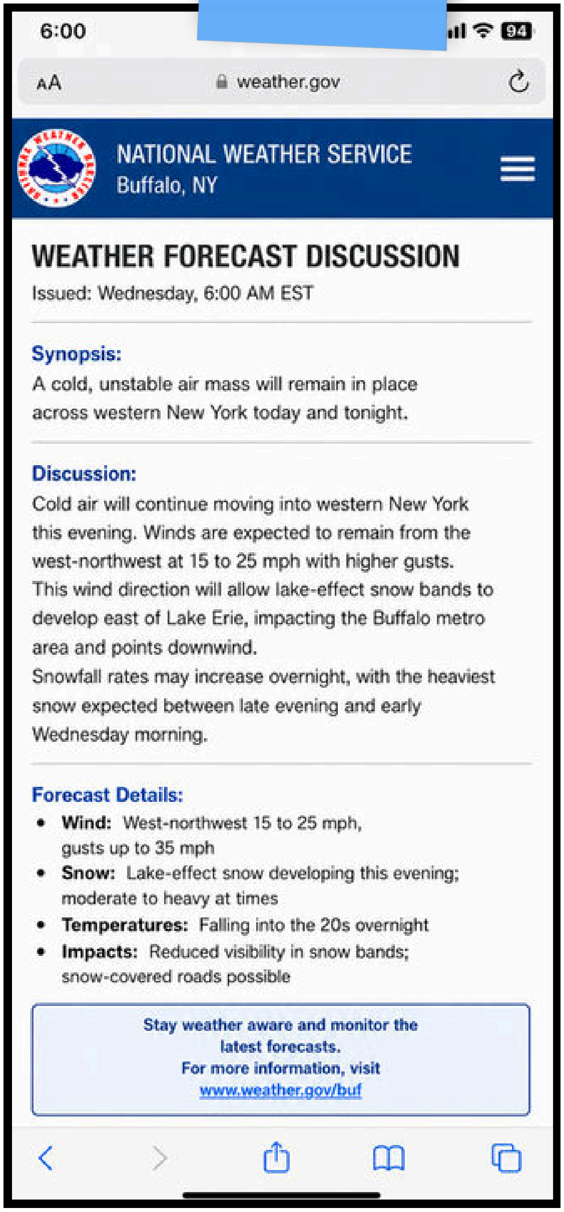
Exhibit C


10:42
The Weather Channel
Buffalo, NY
Radar
Rain
Mix
Snow
mapbox
Lake Ontario
NIAGARA FALLS
90
BUFFALO
Lake Erie
DUNKIRK
86
ERIE
10:35 AM
Now
Radar Past & Future
See All
Today
Hourly
10 Day
Radar
Video
weather.com

Rain moving through

Periods of rain will continue for the next few hours. Drive carefully.

Exhibit D


6:00
94
AA
weather.gov

NATIONAL WEATHER SERVICE
Buffalo, NY

WEATHER FORECAST DISCUSSION

Issued: Wednesday, 6:00 AM EST

Synopsis:

A cold, unstable air mass will remain in place across western New York today and tonight.

Discussion:

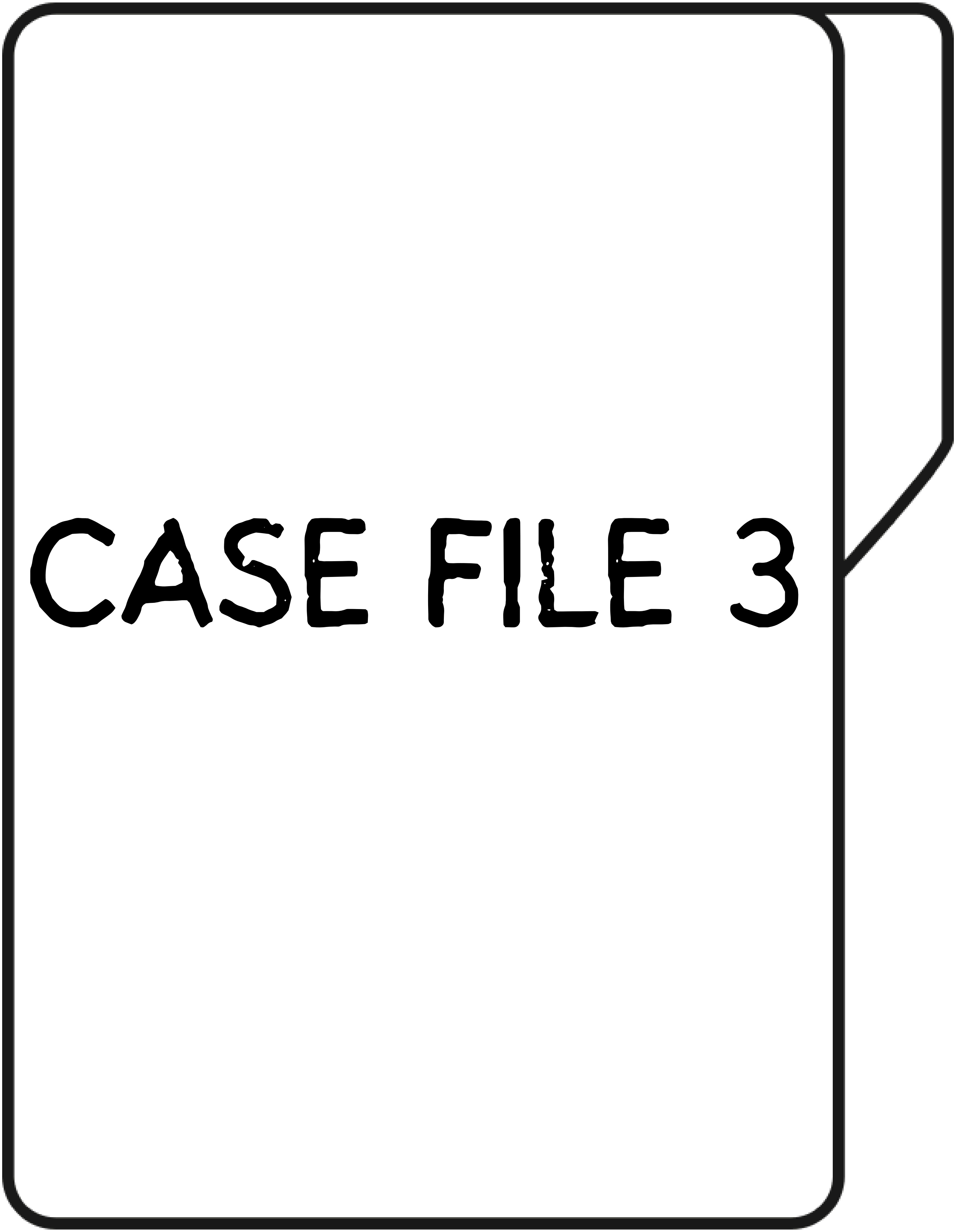
Cold air will continue moving into western New York this evening. Winds are expected to remain from the west-northwest at 15 to 25 mph with higher gusts. This wind direction will allow lake-effect snow bands to develop east of Lake Erie, impacting the Buffalo metro area and points downwind.

Snowfall rates may increase overnight, with the heaviest snow expected between late evening and early Wednesday morning.

Forecast Details:

- **Wind:** West-northwest 15 to 25 mph, gusts up to 35 mph
- **Snow:** Lake-effect snow developing this evening; moderate to heavy at times
- **Temperatures:** Falling into the 20s overnight
- **Impacts:** Reduced visibility in snow bands; snow-covered roads possible

Stay weather aware and monitor the latest forecasts.
For more information, visit www.weather.gov/buf



CASE FILE 3

CASE FILE 3

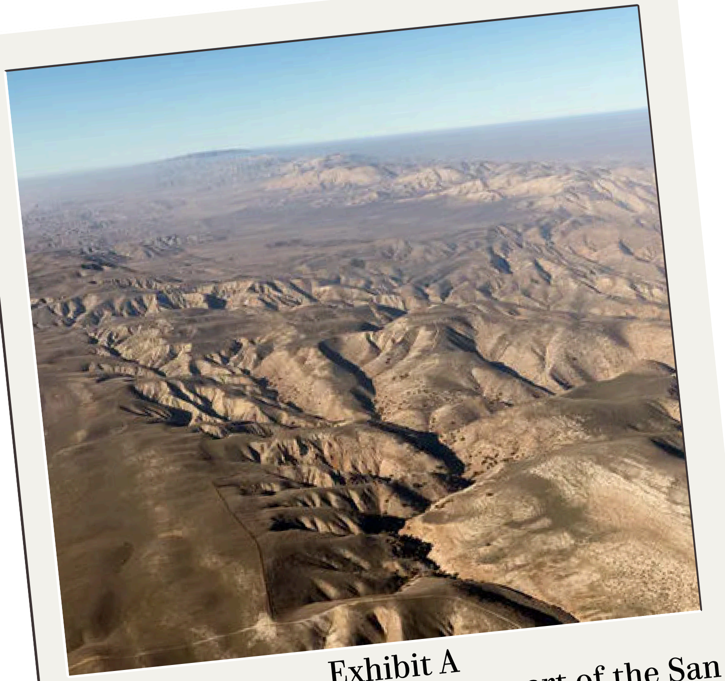


Exhibit A

This aerial photograph shows part of the San Andreas Fault in California. Some sections of the fault can be observed at Earth's surface, but much of the fault extends underground.

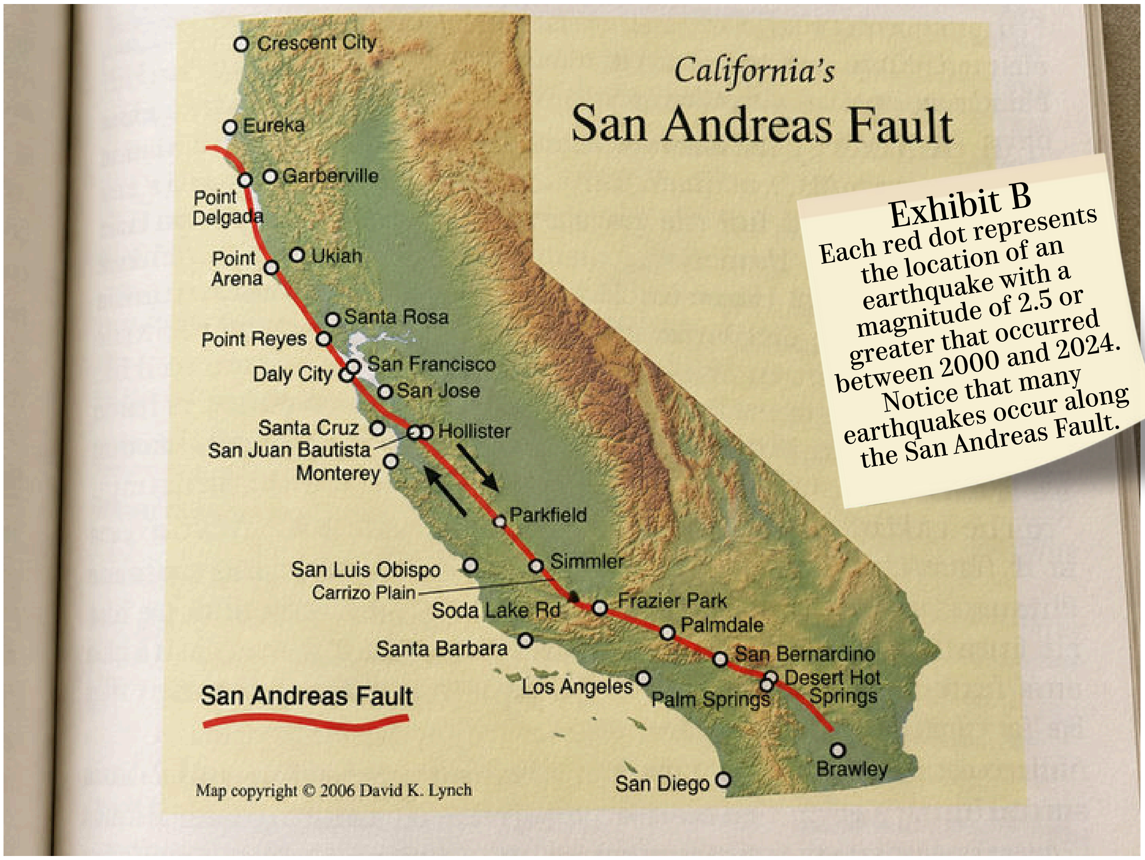

California's
San Andreas Fault
Crescent City
Eureka
Garberville
Point Delgada
Point Arena
Ukiah
Santa Rosa
Point Reyes
Daly City
San Francisco
San Jose
Santa Cruz
Hollister
San Juan Bautista
Monterey
Parkfield
San Luis Obispo
Simmler
Carrizo Plain
Soda Lake Rd
Frazier Park
Palmdale
Santa Barbara
San Bernardino
Los Angeles
Palm Springs
Desert Hot Springs
Brawley
San Diego
San Andreas Fault
Map copyright © 2006 David K. Lynch

Exhibit B

Each red dot represents the location of an earthquake with a magnitude of 2.5 or greater that occurred between 2000 and 2024. Notice that many earthquakes occur along the San Andreas Fault.

CASE FILE 3

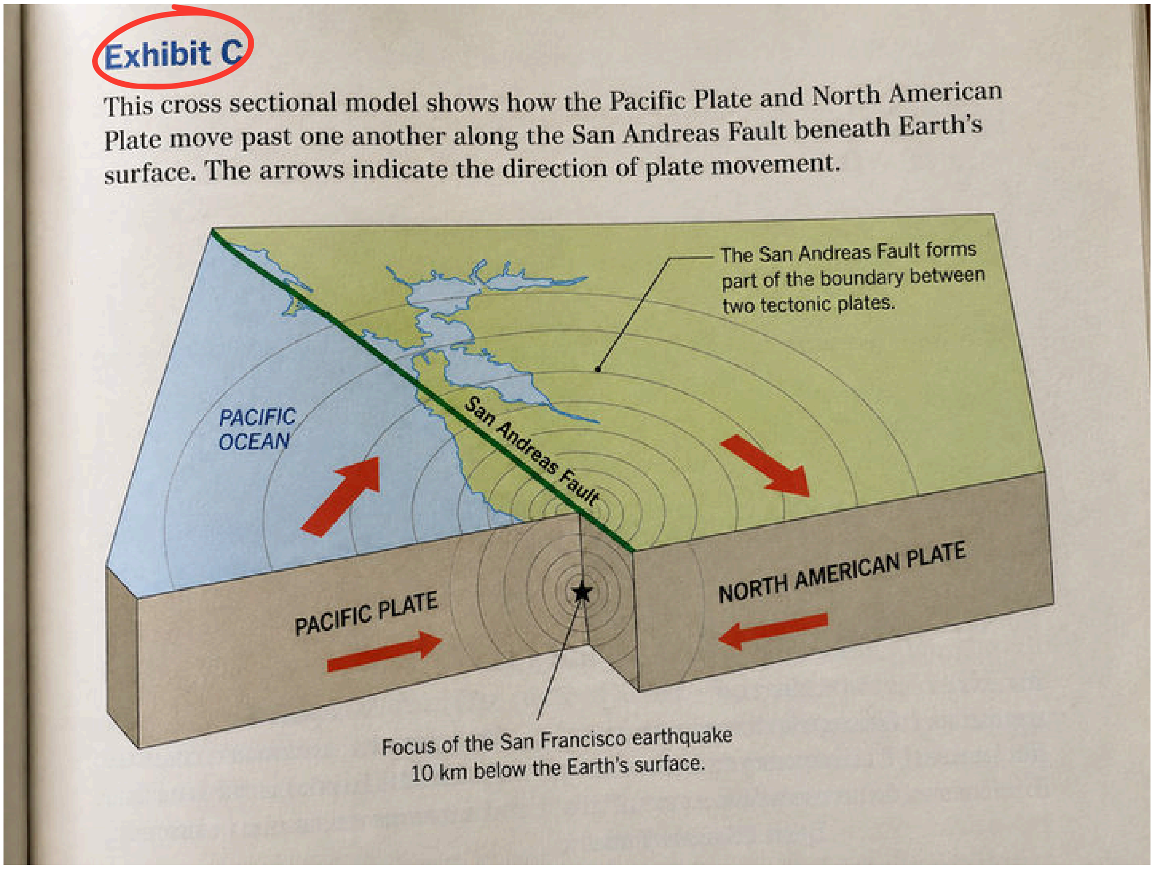


Exhibit C

This cross sectional model shows how the Pacific Plate and North American Plate move past one another along the San Andreas Fault beneath Earth's surface. The arrows indicate the direction of plate movement.

The San Andreas Fault forms part of the boundary between two tectonic plates.
PACIFIC OCEAN
San Andreas Fault
PACIFIC PLATE
NORTH AMERICAN PLATE
Focus of the San Francisco earthquake 10 km below the Earth's surface.

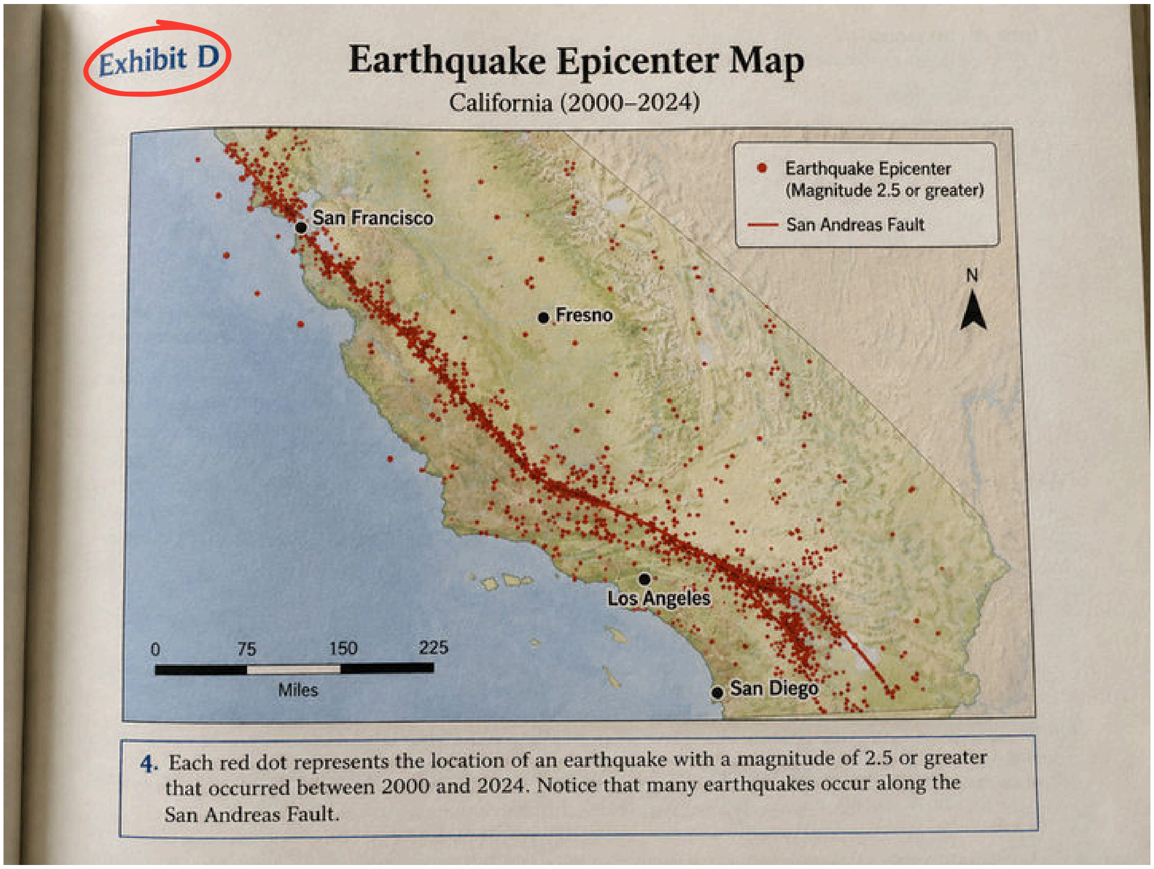


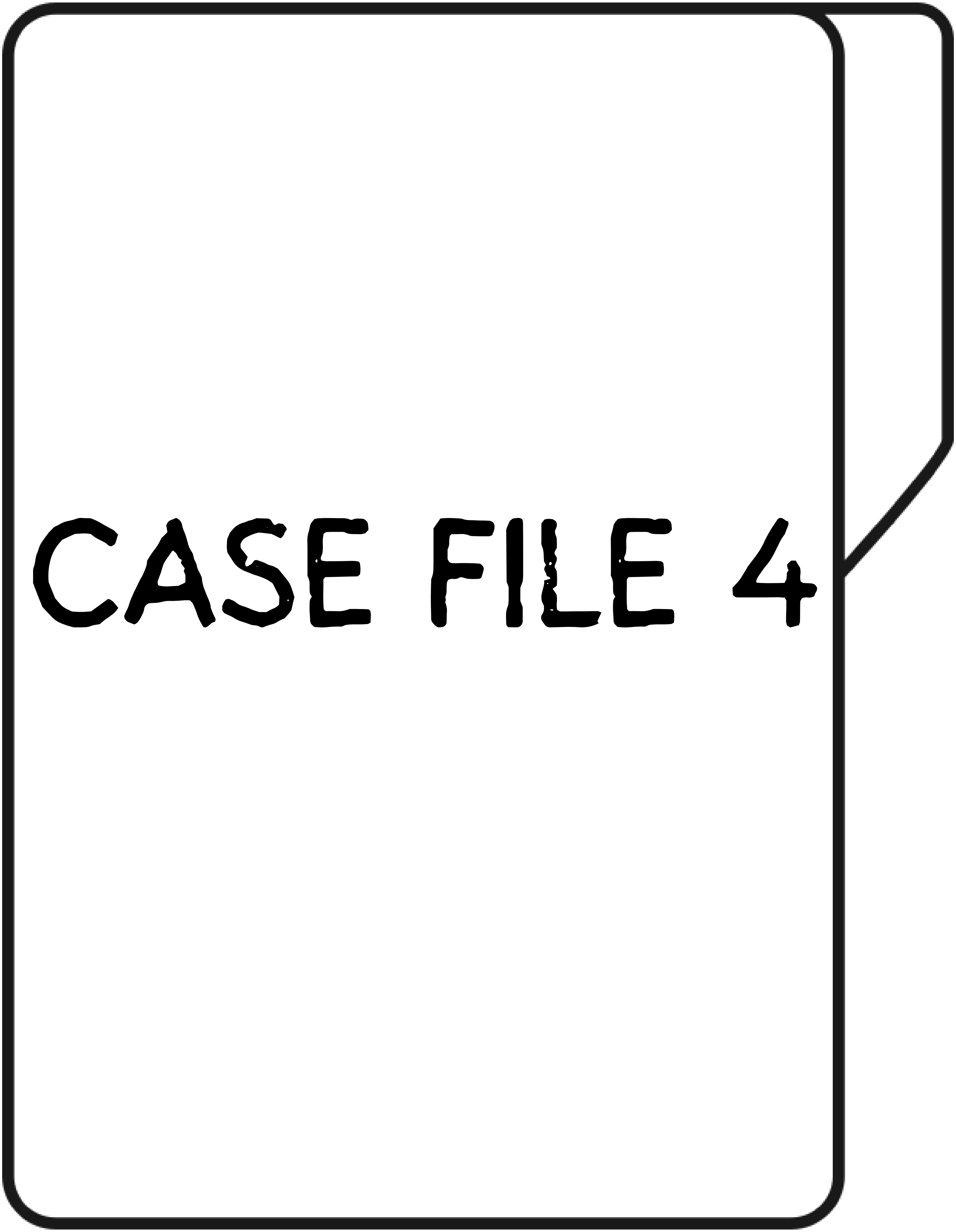
Exhibit D

Earthquake Epicenter Map

California (2000–2024)

● Earthquake Epicenter (Magnitude 2.5 or greater)
— San Andreas Fault
San Francisco
Fresno
Los Angeles
San Diego
N
0
75
150
225
Miles

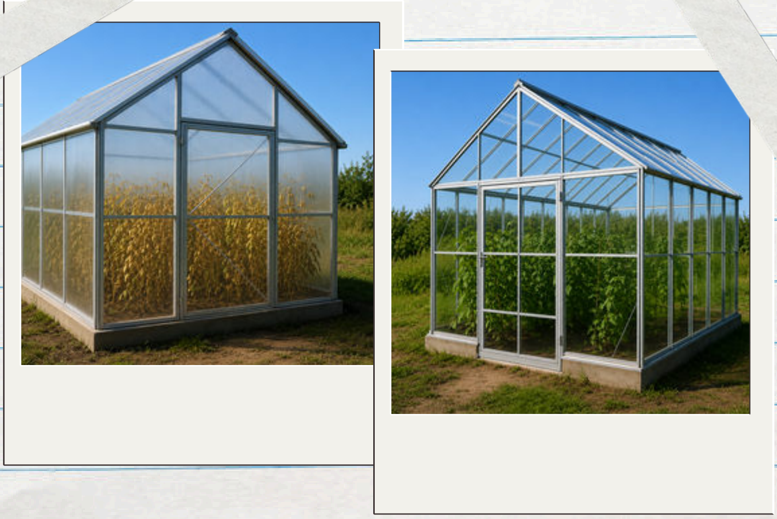
4. Each red dot represents the location of an earthquake with a magnitude of 2.5 or greater that occurred between 2000 and 2024. Notice that many earthquakes occur along the San Andreas Fault.



CASE FILE 4

CASE FILE 4

Exhibit A



Scientists noticed that plants in one greenhouse are healthy, while plants in another greenhouse are showing signs of poor growth.

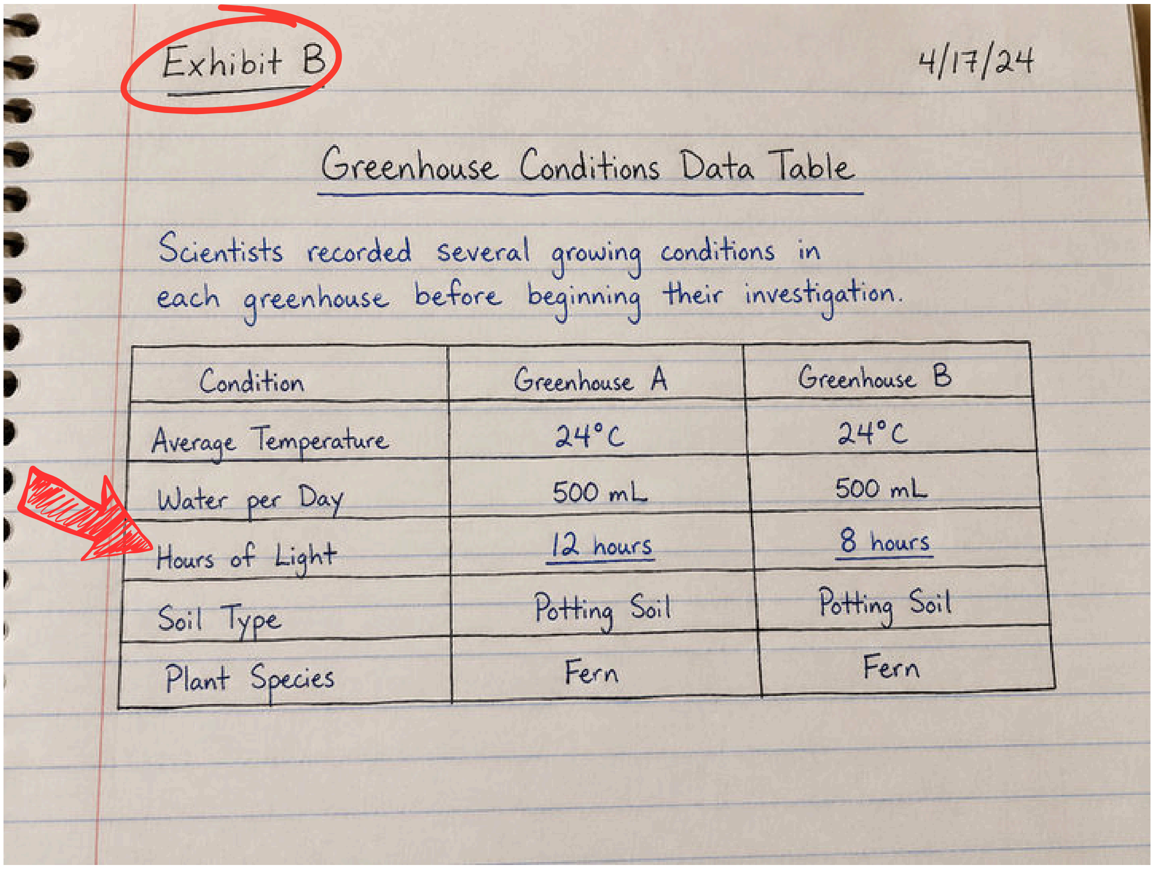


Exhibit B

4/17/24

Greenhouse Conditions Data Table

Scientists recorded several growing conditions in each greenhouse before beginning their investigation.

Condition	Greenhouse A	Greenhouse B
Average Temperature	24°C	24°C
Water per Day	500 mL	500 mL
Hours of Light	12 hours	8 hours
Soil Type	Potting Soil	Potting Soil
Plant Species	Fern	Fern

CASE FILE 4

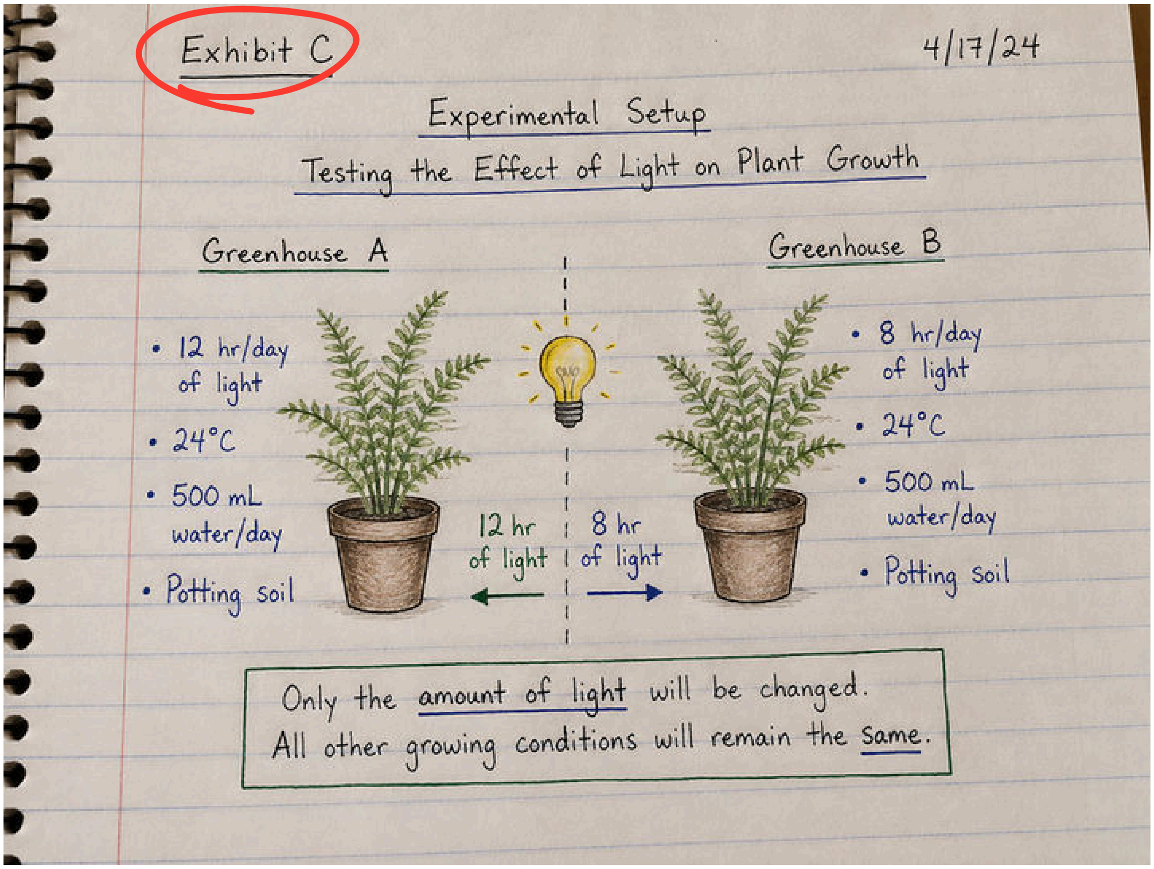


Exhibit C

4/17/24

Experimental Setup

Testing the Effect of Light on Plant Growth

Greenhouse A

- 12 hr/day of light
- 24°C
- 500 mL water/day
- Potting soil

Greenhouse B

- 8 hr/day of light
- 24°C
- 500 mL water/day
- Potting soil

12 hr of light
8 hr of light

Only the amount of light will be changed.
All other growing conditions will remain the same.

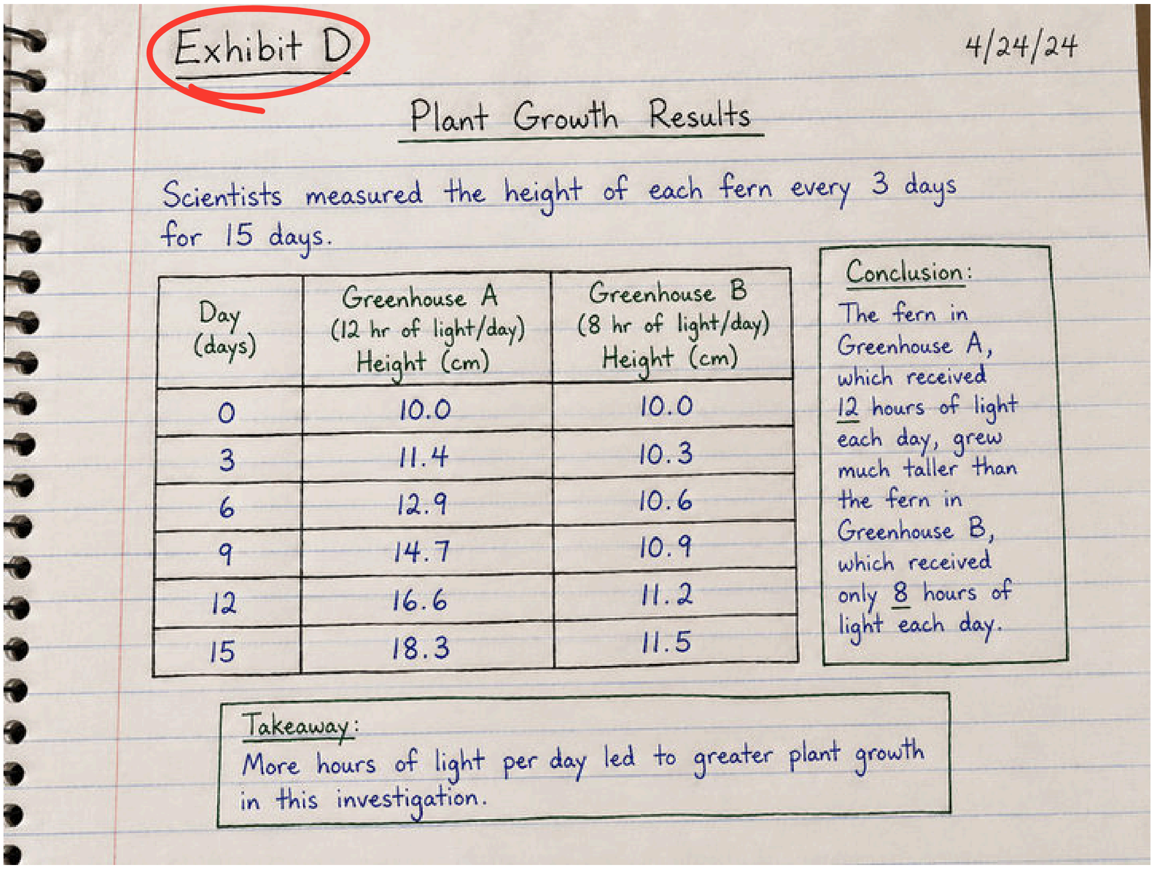


Exhibit D

4/24/24

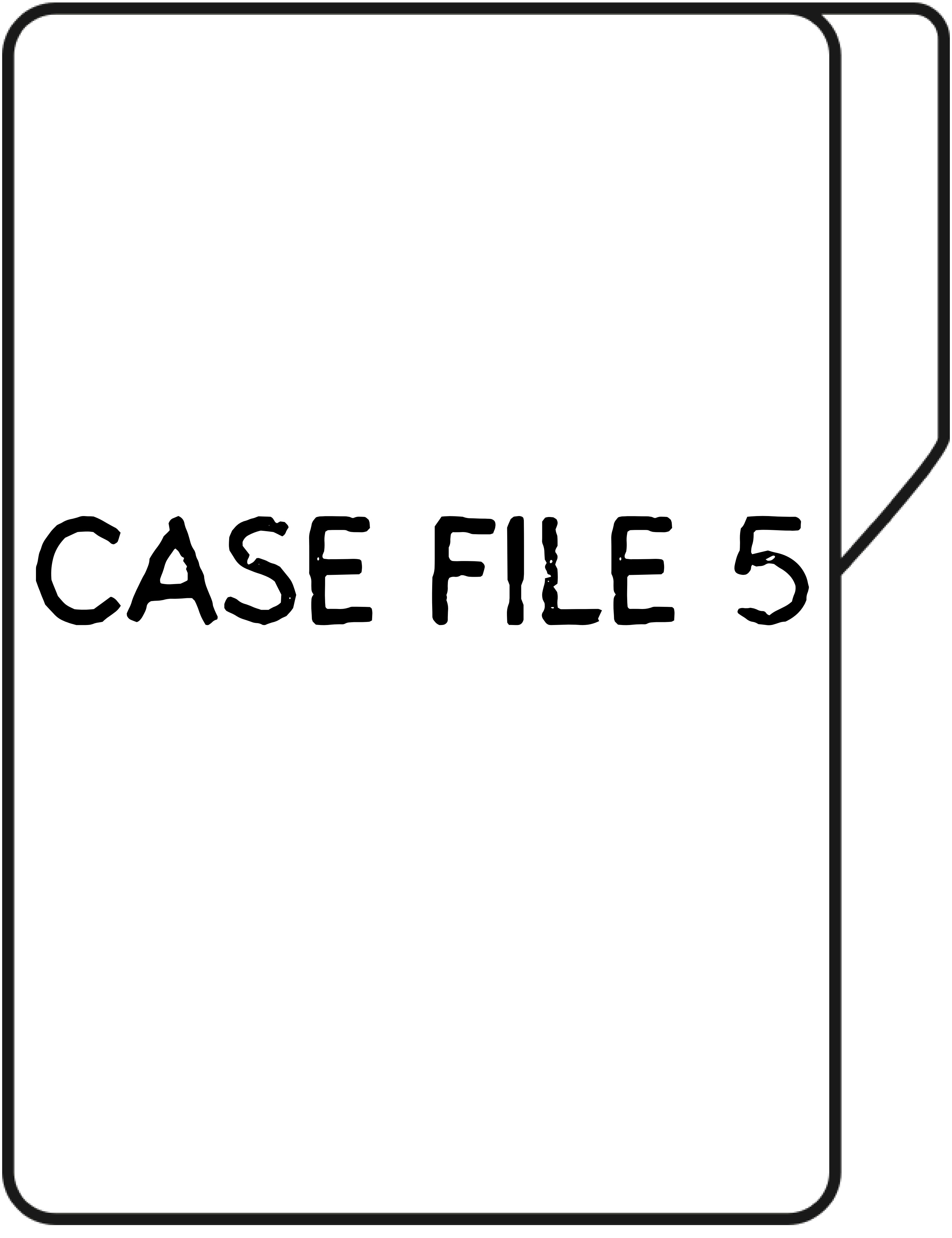
Plant Growth Results

Scientists measured the height of each fern every 3 days for 15 days.

Day (days)	Greenhouse A (12 hr of light/day) Height (cm)	Greenhouse B (8 hr of light/day) Height (cm)
0	10.0	10.0
3	11.4	10.3
6	12.9	10.6
9	14.7	10.9
12	16.6	11.2
15	18.3	11.5

Conclusion:
The fern in Greenhouse A, which received 12 hours of light each day, grew much taller than the fern in Greenhouse B, which received only 8 hours of light each day.

Takeaway:
More hours of light per day led to greater plant growth in this investigation.



CASE FILE 5

CASE FILE 5

Exhibit A: The Lake Erie System

- This image shows several components of the Lake Erie system, including land, water, living things, and human activities. These components can interact with and affect one another.

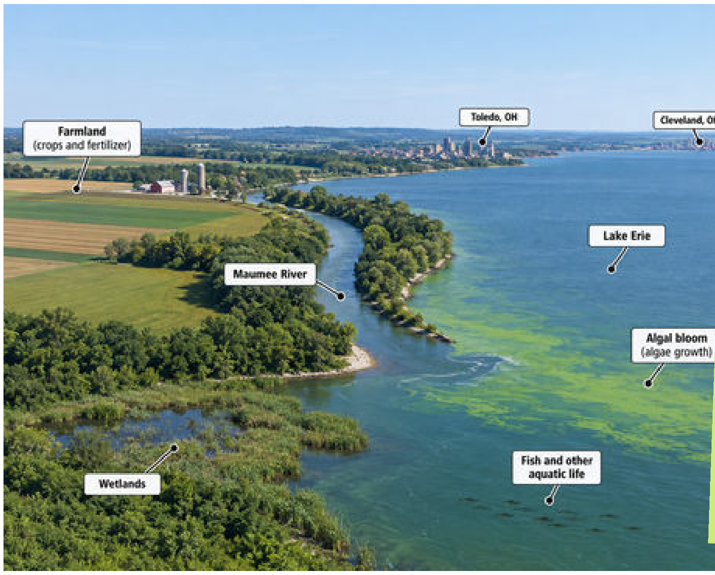
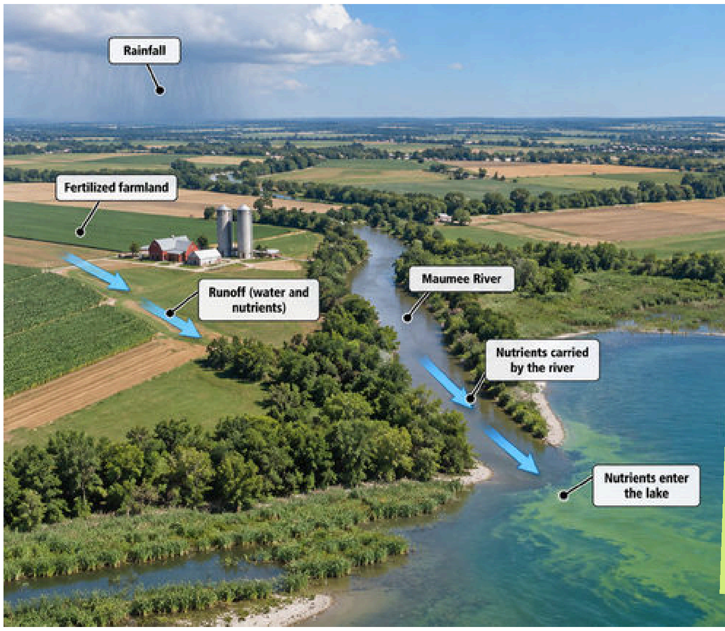

Farmland (crops and fertilizer)
Toledo, OH
Cleveland, OH
Lake Erie
Maumee River
Algal bloom (algae growth)
Fish and other aquatic life
Wetlands

Exhibit B: Nutrients Enter Lake Erie

- Rainfall can wash nutrients from fertilized farmland into nearby waterways. The Maumee River carries this nutrient-rich water into Lake Erie.


Rainfall
Fertilized farmland
Runoff (water and nutrients)
Maumee River
Nutrients carried by the river
Nutrients enter the lake

CASE FILE 5

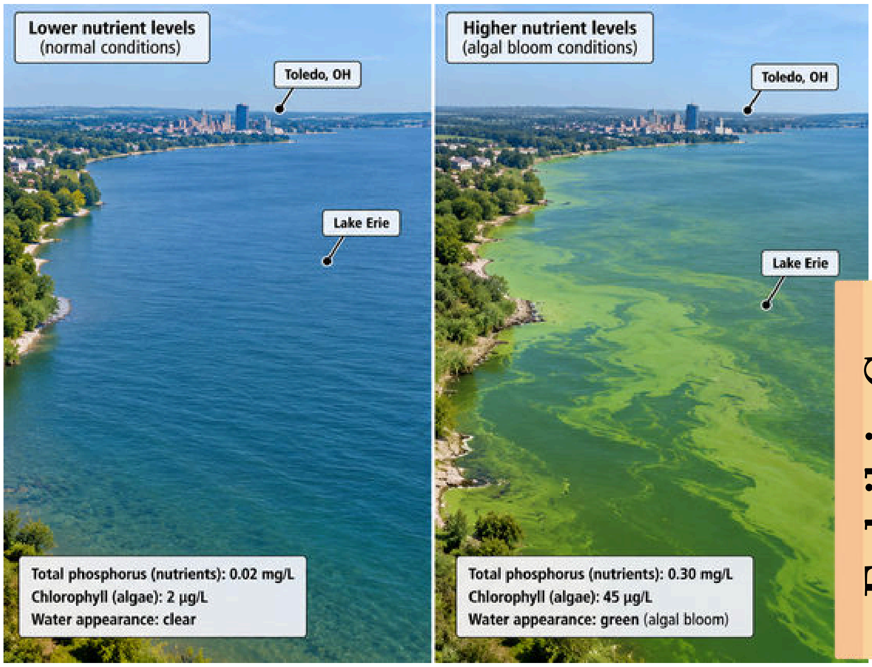
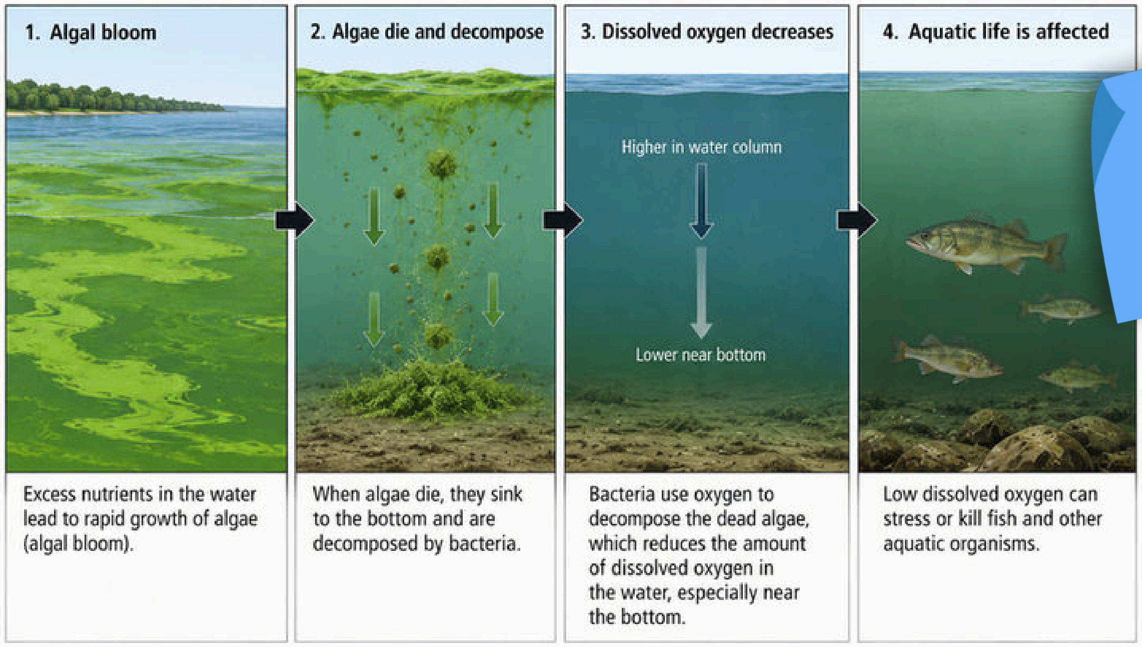

Lower nutrient levels
(normal conditions)
Toledo, OH
Lake Erie
Total phosphorus (nutrients): 0.02 mg/L
Chlorophyll (algae): 2 µg/L
Water appearance: clear
Higher nutrient levels
(algal bloom conditions)
Toledo, OH
Lake Erie
Total phosphorus (nutrients): 0.30 mg/L
Chlorophyll (algae): 45 µg/L
Water appearance: green (algal bloom)

Exhibit C

Scientists compared areas of Lake Erie with different nutrient levels. Areas with higher nutrient levels had more algae and greener water..


1. Algal bloom
2. Algae die and decompose
3. Dissolved oxygen decreases
Higher in water column
Lower near bottom
4. Aquatic life is affected

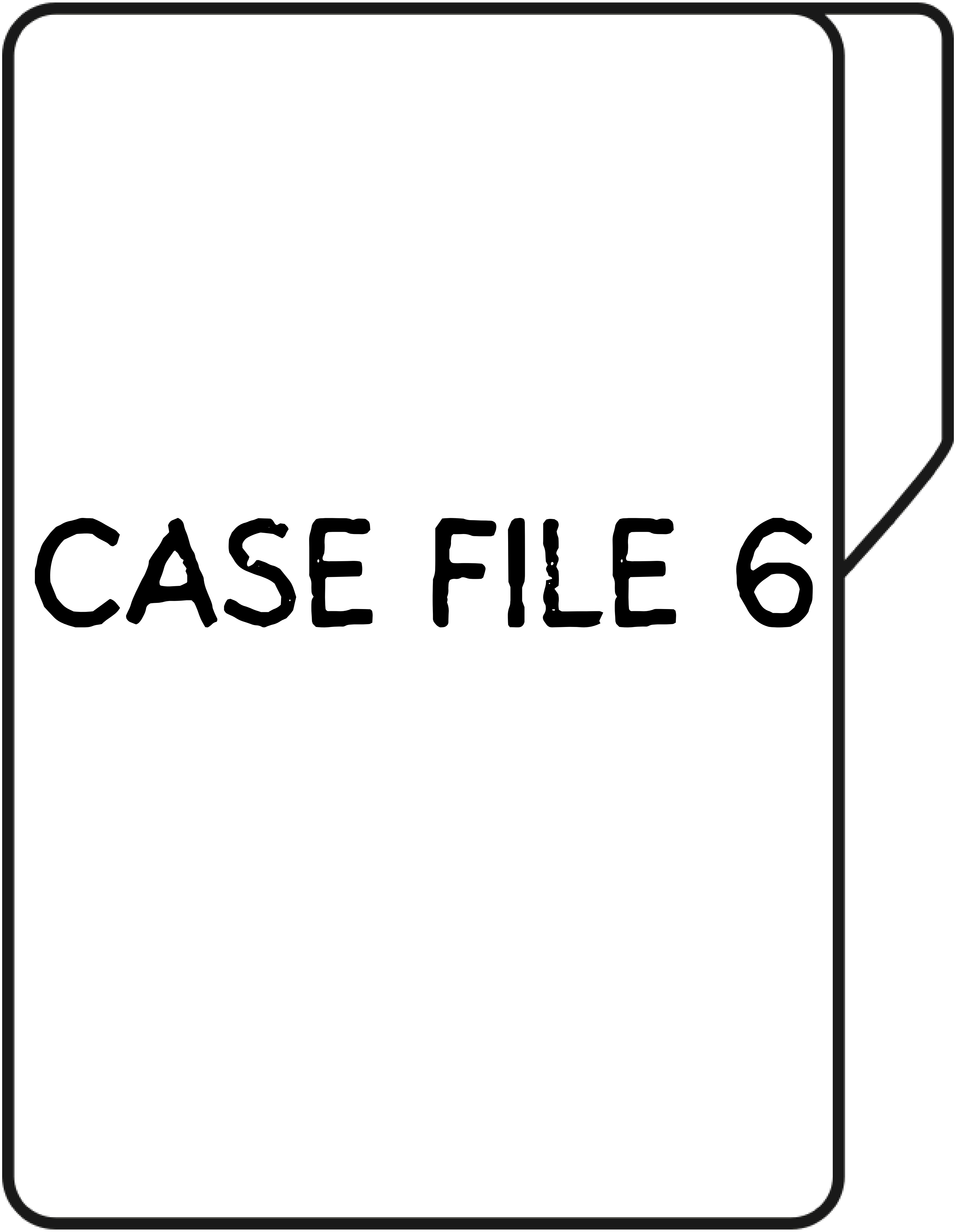
1. Algal bloom: Excess nutrients in the water lead to rapid growth of algae (algal bloom).

2. Algae die and decompose: When algae die, they sink to the bottom and are decomposed by bacteria.

3. Dissolved oxygen decreases: Bacteria use oxygen to decompose the dead algae, which reduces the amount of dissolved oxygen in the water, especially near the bottom.

4. Aquatic life is affected: Low dissolved oxygen can stress or kill fish and other aquatic organisms.

Exhibit D

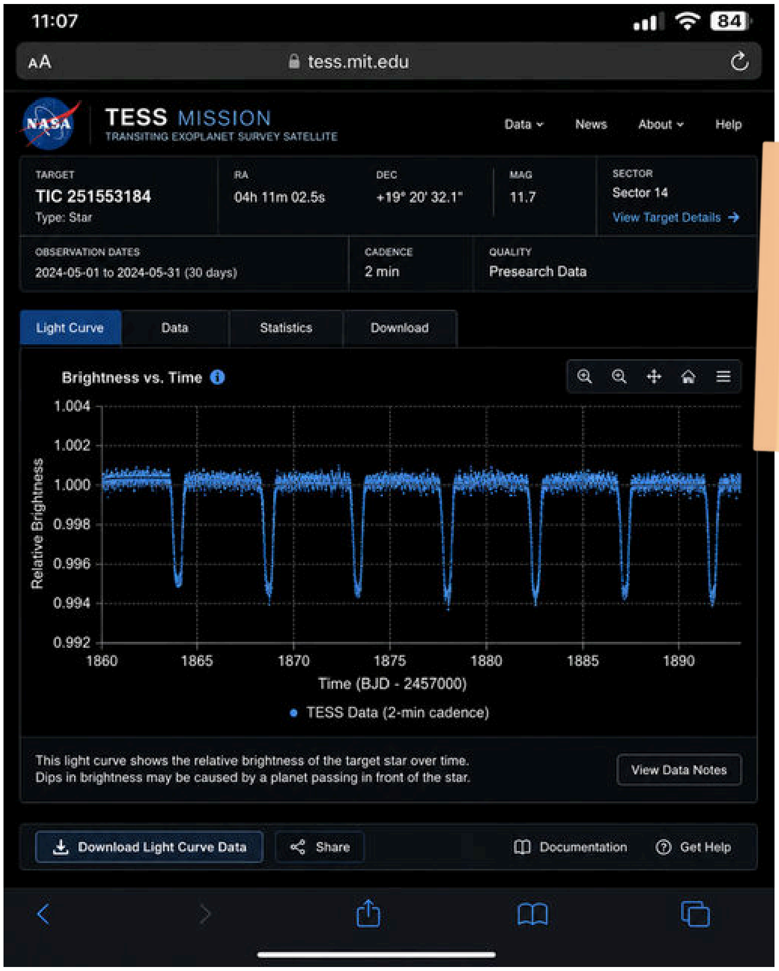


CASE FILE 6

CASE FILE 6

Exhibit A

Brightness measurements collected by NASA's Transiting Exoplanet Survey Satellite (TESS). The graph shows how the brightness of a distant star changed during a 30-day observation period.


11:07
84
AA
tess.mit.edu
NASA
Data
News
About
Help
Light Curve
Data
Statistics
Download
Brightness vs. Time
1.004
1.002
1.000
0.998
0.996
0.994
0.992
Relative Brightness
1860
1865
1870
1875
1880
1885
1890
Time (BJD - 2457000)
TESS Data (2-min cadence)
View Data Notes
Download Light Curve Data
Share
Documentation
Get Help

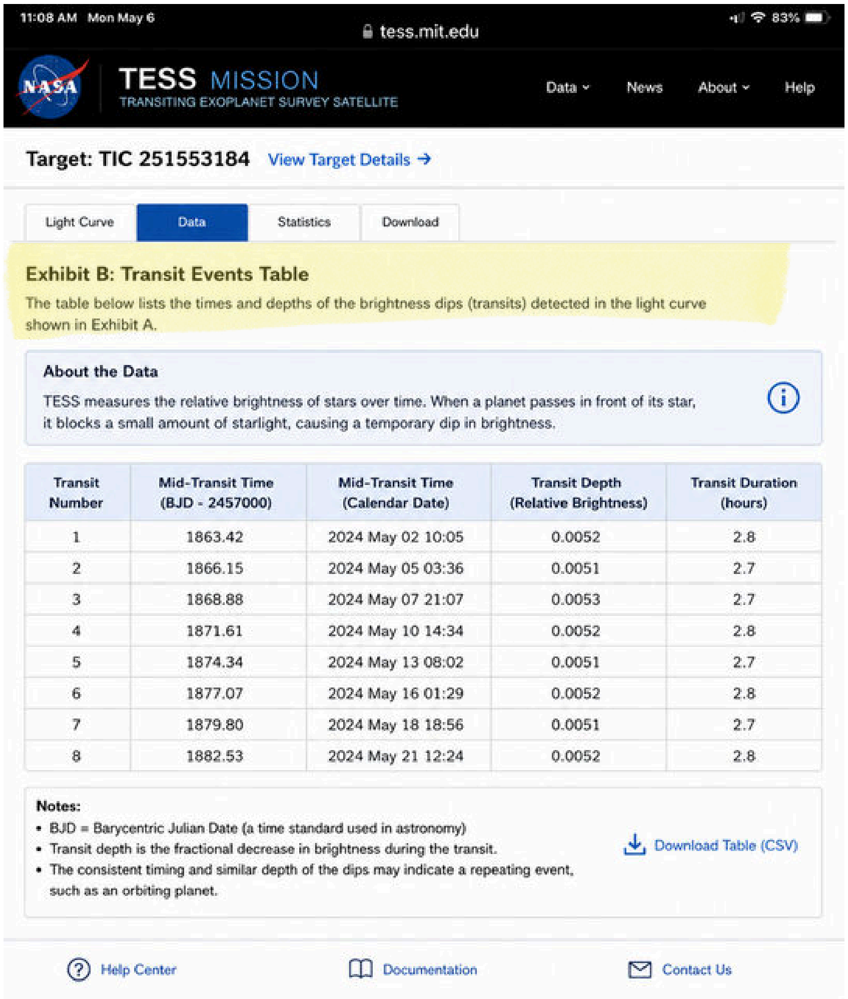
TESS MISSION

TRANSITING EXOPLANET SURVEY SATELLITE

TARGET	RA	DEC	MAG	SECTOR
TIC 251553184 Type: Star	04h 11m 02.5s	+19° 20' 32.1"	11.7	Sector 14 View Target Details →

OBSERVATION DATES	CADENCE	QUALITY
2024-05-01 to 2024-05-31 (30 days)	2 min	Prereasearch Data

This light curve shows the relative brightness of the target star over time. Dips in brightness may be caused by a planet passing in front of the star.


11:08 AM Mon May 6
83%
tess.mit.edu
NASA
Data
News
About
Help
Light Curve
Data
Statistics
Download
Download Table (CSV)
Help Center
Documentation
Contact Us

TESS MISSION

TRANSITING EXOPLANET SURVEY SATELLITE

Target: TIC 251553184 View Target Details →

Exhibit B: Transit Events Table

The table below lists the times and depths of the brightness dips (transits) detected in the light curve shown in Exhibit A.

About the Data

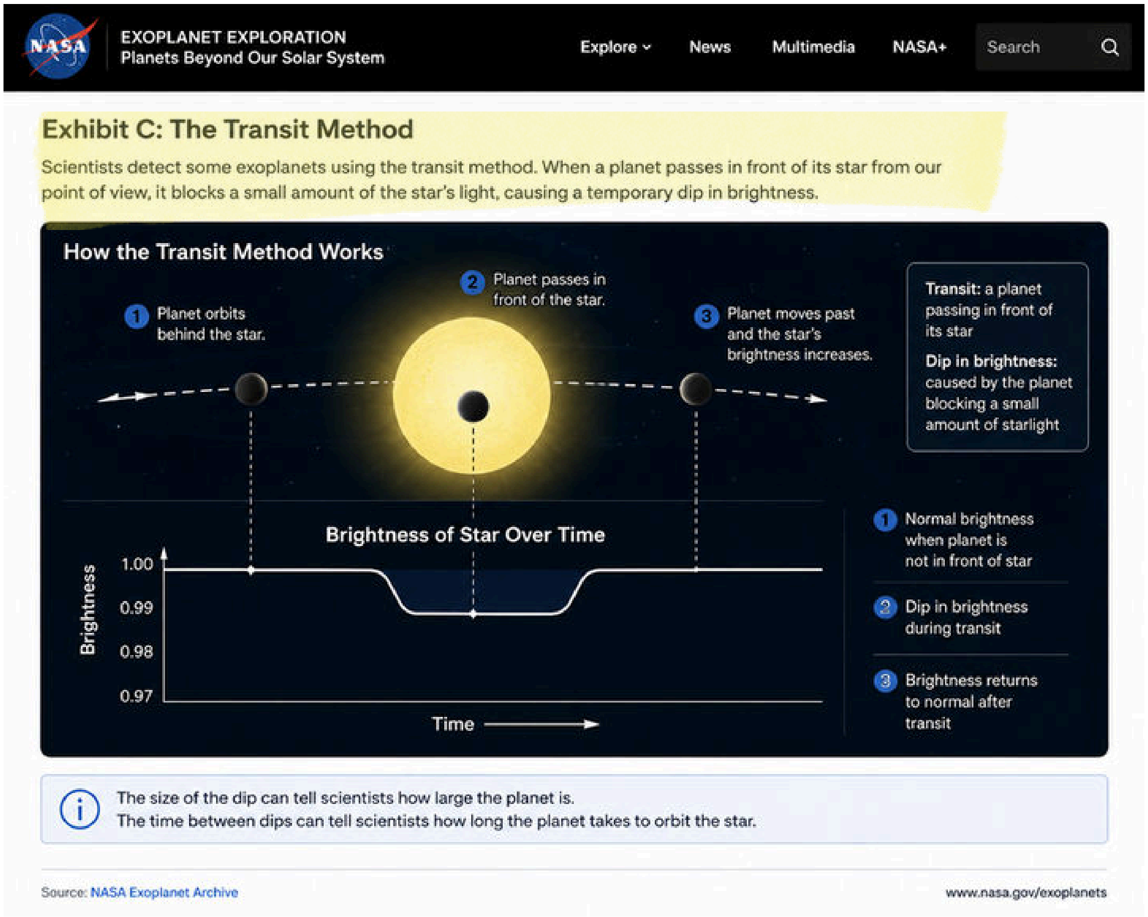
TESS measures the relative brightness of stars over time. When a planet passes in front of its star, it blocks a small amount of starlight, causing a temporary dip in brightness.

Transit Number	Mid-Transit Time (BJD - 2457000)	Mid-Transit Time (Calendar Date)	Transit Depth (Relative Brightness)	Transit Duration (hours)
1	1863.42	2024 May 02 10:05	0.0052	2.8
2	1866.15	2024 May 05 03:36	0.0051	2.7
3	1868.88	2024 May 07 21:07	0.0053	2.7
4	1871.61	2024 May 10 14:34	0.0052	2.8
5	1874.34	2024 May 13 08:02	0.0051	2.7
6	1877.07	2024 May 16 01:29	0.0052	2.8
7	1879.80	2024 May 18 18:56	0.0051	2.7
8	1882.53	2024 May 21 12:24	0.0052	2.8

Notes:

- BJD = Barycentric Julian Date (a time standard used in astronomy)
- Transit depth is the fractional decrease in brightness during the transit.
- The consistent timing and similar depth of the dips may indicate a repeating event, such as an orbiting planet.

CASE FILE 6



EXOPLANET EXPLORATION
Planets Beyond Our Solar System

Exhibit C: The Transit Method

Scientists detect some exoplanets using the transit method. When a planet passes in front of its star from our point of view, it blocks a small amount of the star's light, causing a temporary dip in brightness.

How the Transit Method Works

① Planet orbits behind the star.

② Planet passes in front of the star.

③ Planet moves past and the star's brightness increases.

Transit: a planet passing in front of its star

Dip in brightness: caused by the planet blocking a small amount of starlight

Brightness of Star Over Time

① Normal brightness when planet is not in front of star

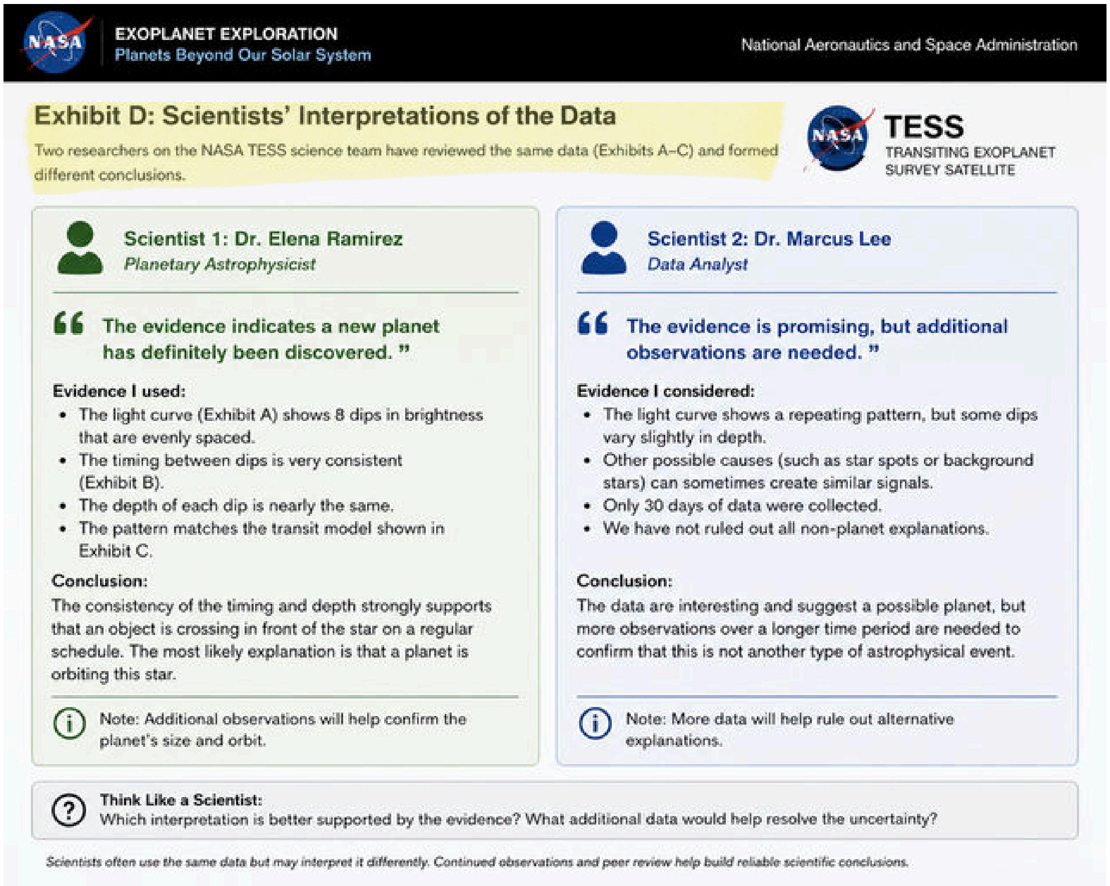
② Dip in brightness during transit

③ Brightness returns to normal after transit

The size of the dip can tell scientists how large the planet is.
The time between dips can tell scientists how long the planet takes to orbit the star.

Source: NASA Exoplanet Archive

www.nasa.gov/exoplanets



EXOPLANET EXPLORATION
Planets Beyond Our Solar System

National Aeronautics and Space Administration

Exhibit D: Scientists' Interpretations of the Data

Two researchers on the NASA TESS science team have reviewed the same data (Exhibits A–C) and formed different conclusions.

TESS
TRANSITING EXOPLANET SURVEY SATELLITE

Scientist 1: Dr. Elena Ramirez

Planetary Astrophysicist

> **“The evidence indicates a new planet has definitely been discovered.”**

Evidence I used:

- The light curve (Exhibit A) shows 8 dips in brightness that are evenly spaced.
- The timing between dips is very consistent (Exhibit B).
- The depth of each dip is nearly the same.
- The pattern matches the transit model shown in Exhibit C.

Conclusion:

The consistency of the timing and depth strongly supports that an object is crossing in front of the star on a regular schedule. The most likely explanation is that a planet is orbiting this star.

Note: Additional observations will help confirm the planet's size and orbit.

Scientist 2: Dr. Marcus Lee

Data Analyst

> **“The evidence is promising, but additional observations are needed.”**

Evidence I considered:

- The light curve shows a repeating pattern, but some dips vary slightly in depth.
- Other possible causes (such as star spots or background stars) can sometimes create similar signals.
- Only 30 days of data were collected.
- We have not ruled out all non-planet explanations.

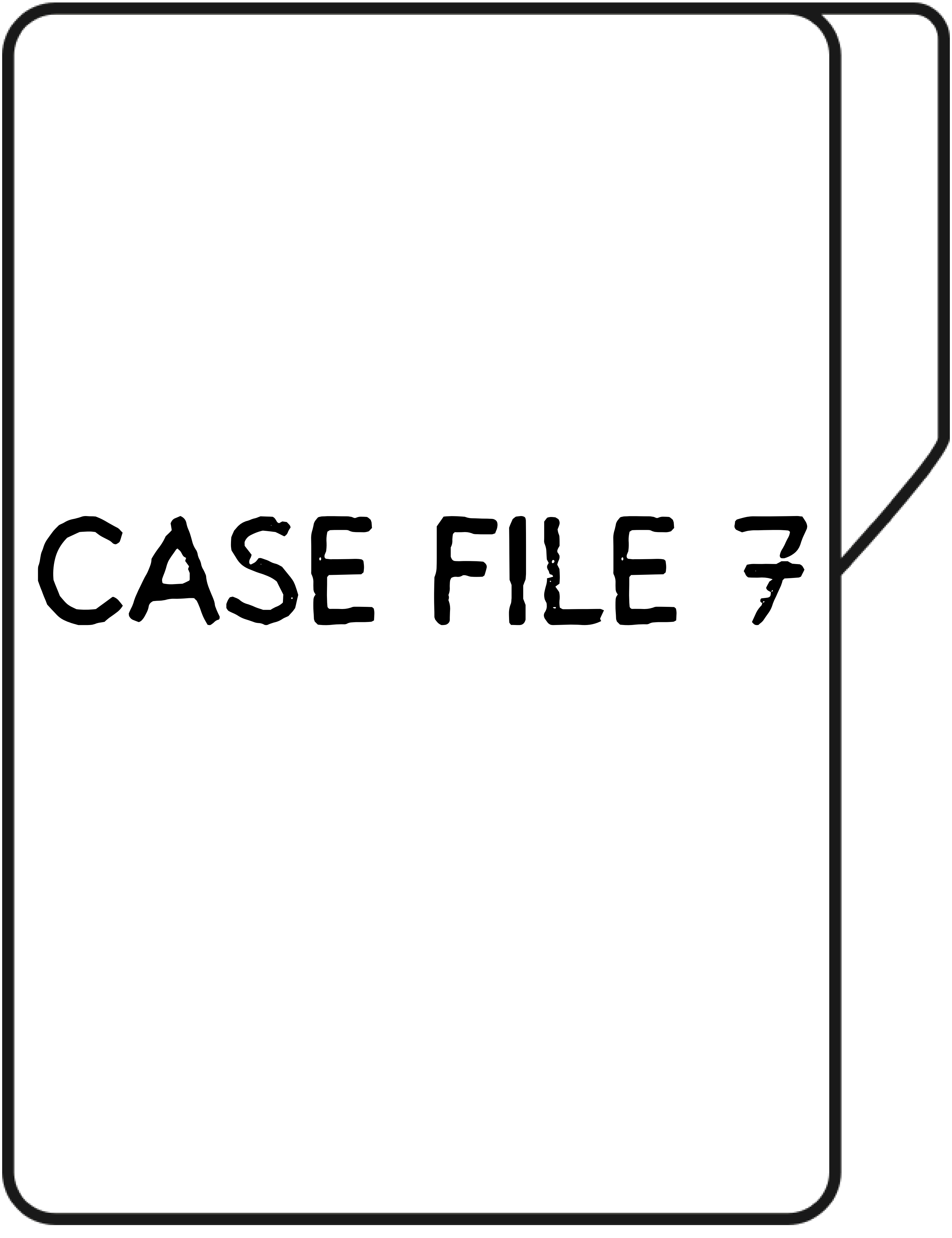
Conclusion:

The data are interesting and suggest a possible planet, but more observations over a longer time period are needed to confirm that this is not another type of astrophysical event.

Note: More data will help rule out alternative explanations.

Think Like a Scientist:
Which interpretation is better supported by the evidence? What additional data would help resolve the uncertainty?

Scientists often use the same data but may interpret it differently. Continued observations and peer review help build reliable scientific conclusions.



CASE FILE 7

CASE FILE 7

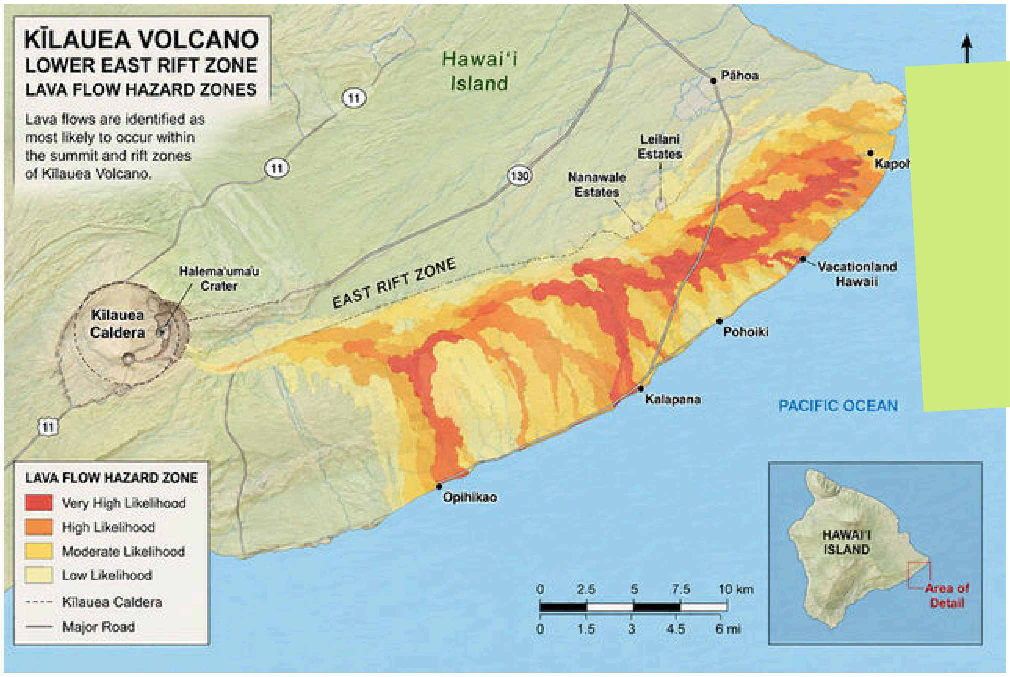

KĪLAUEA VOLCANO
LOWER EAST RIFT ZONE
LAVA FLOW HAZARD ZONES
Lava flows are identified as most likely to occur within the summit and rift zones of Kīlauea Volcano.
Hawai'i Island
Pāhoa
Leilani Estates
Nanawale Estates
Kapoh
Vacationland Hawaii
Pohoiki
Kalapana
Opihikao
Halema'uma'u Crater
Kīlauea Caldera
EAST RIFT ZONE
PACIFIC OCEAN
11
130
LAVA FLOW HAZARD ZONE
🟥 Very High Likelihood
🟧 High Likelihood
🟨 Moderate Likelihood
▭ Low Likelihood
---- Kīlauea Caldera
— Major Road
0 2.5 5 7.5 10 km
0 1.5 3 4.5 6 mi
HAWAI'I ISLAND
Area of Detail

Exhibit A

Map showing lava flow hazard zones around Kīlauea Volcano.

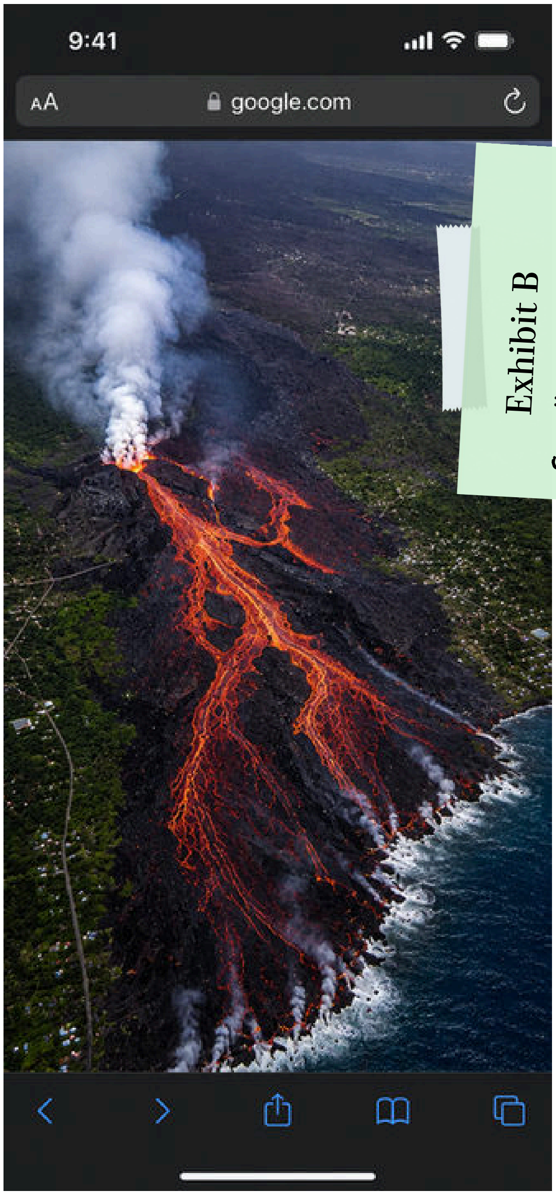
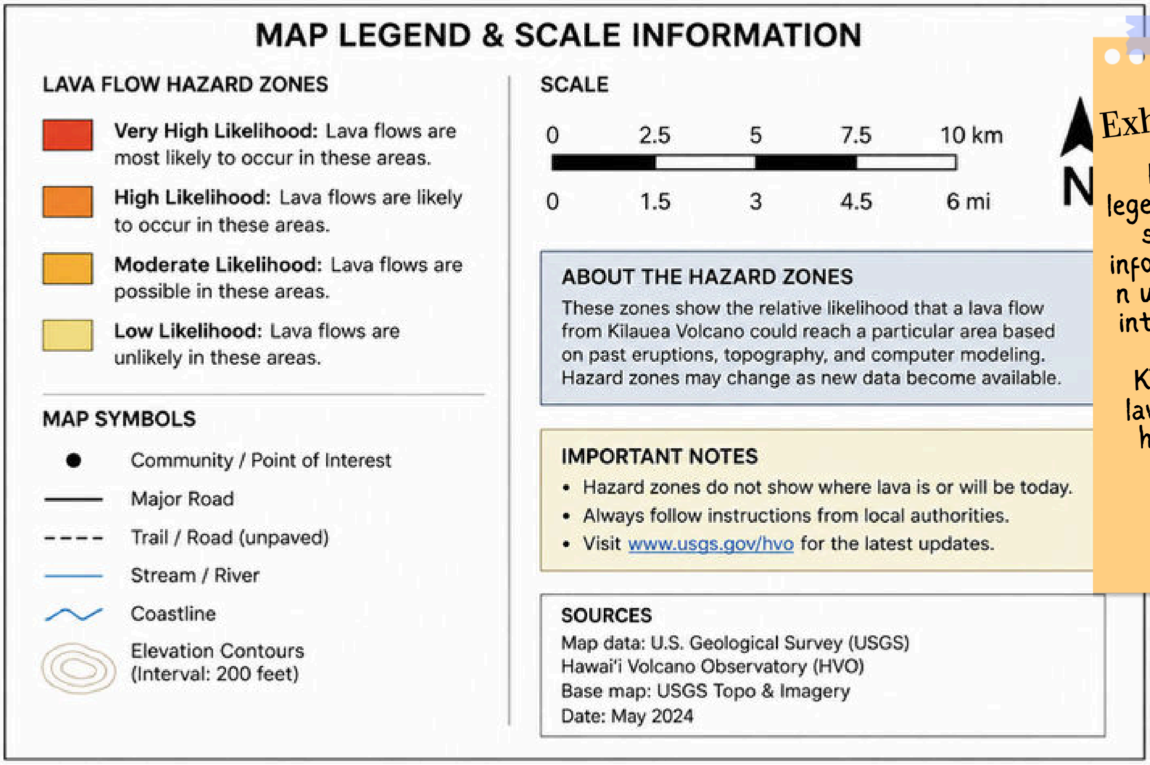

9:41
AA
google.com

Exhibit B

Satellite image showing an active lava flow moving across the landscape toward the coast.

CASE FILE 7



MAP LEGEND & SCALE INFORMATION

LAVA FLOW HAZARD ZONES

- 🟥 **Very High Likelihood:** Lava flows are most likely to occur in these areas.
- 🟧 **High Likelihood:** Lava flows are likely to occur in these areas.
- 🟨 **Moderate Likelihood:** Lava flows are possible in these areas.
- 🟨 **Low Likelihood:** Lava flows are unlikely in these areas.

MAP SYMBOLS

- ● Community / Point of Interest
- — Major Road
- - - - Trail / Road (unpaved)
- Stream / River
- Coastline
- Elevation Contours (Interval: 200 feet)

SCALE

0 2.5 5 7.5 10 km

0 1.5 3 4.5 6 mi

N

ABOUT THE HAZARD ZONES

These zones show the relative likelihood that a lava flow from Kilauea Volcano could reach a particular area based on past eruptions, topography, and computer modeling. Hazard zones may change as new data become available.

IMPORTANT NOTES

- Hazard zones do not show where lava is or will be today.
- Always follow instructions from local authorities.
- Visit www.usgs.gov/hvo for the latest updates.

SOURCES

Map data: U.S. Geological Survey (USGS)
Hawai'i Volcano Observatory (HVO)
Base map: USGS Topo & Imagery
Date: May 2024

Exhibit C

Map legend and scale information used to interpret the Kīlauea lava flow hazard map.

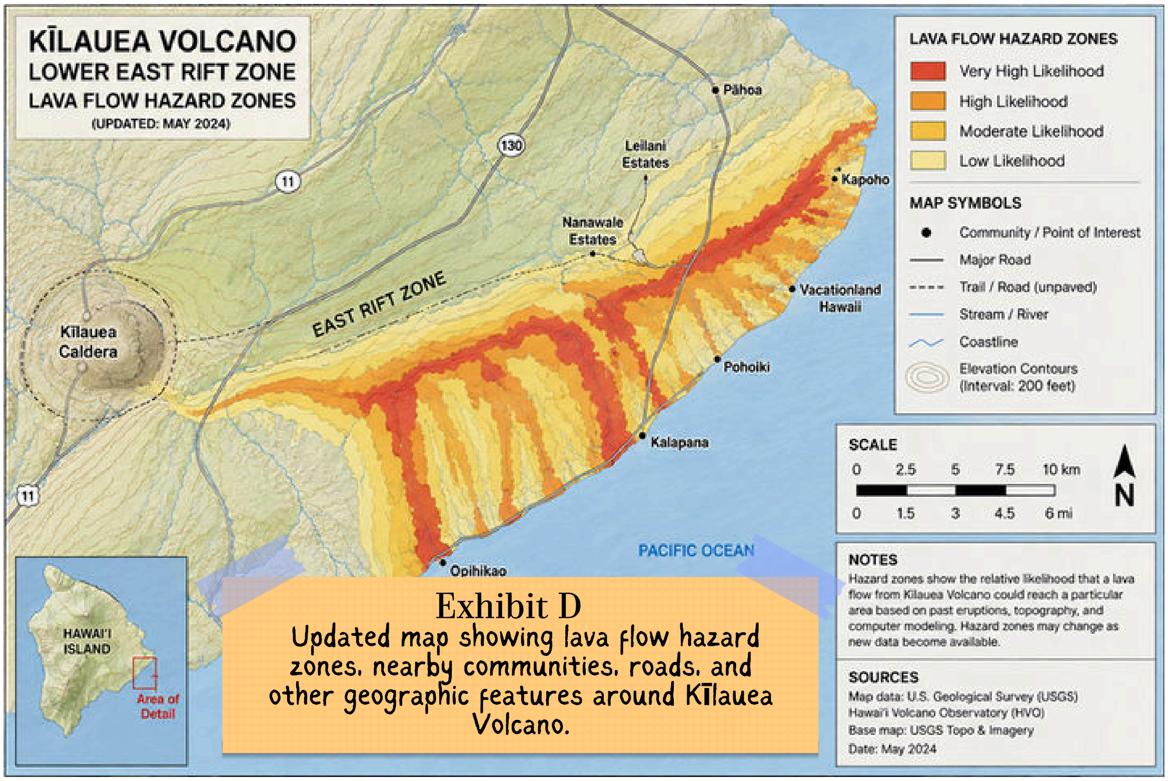
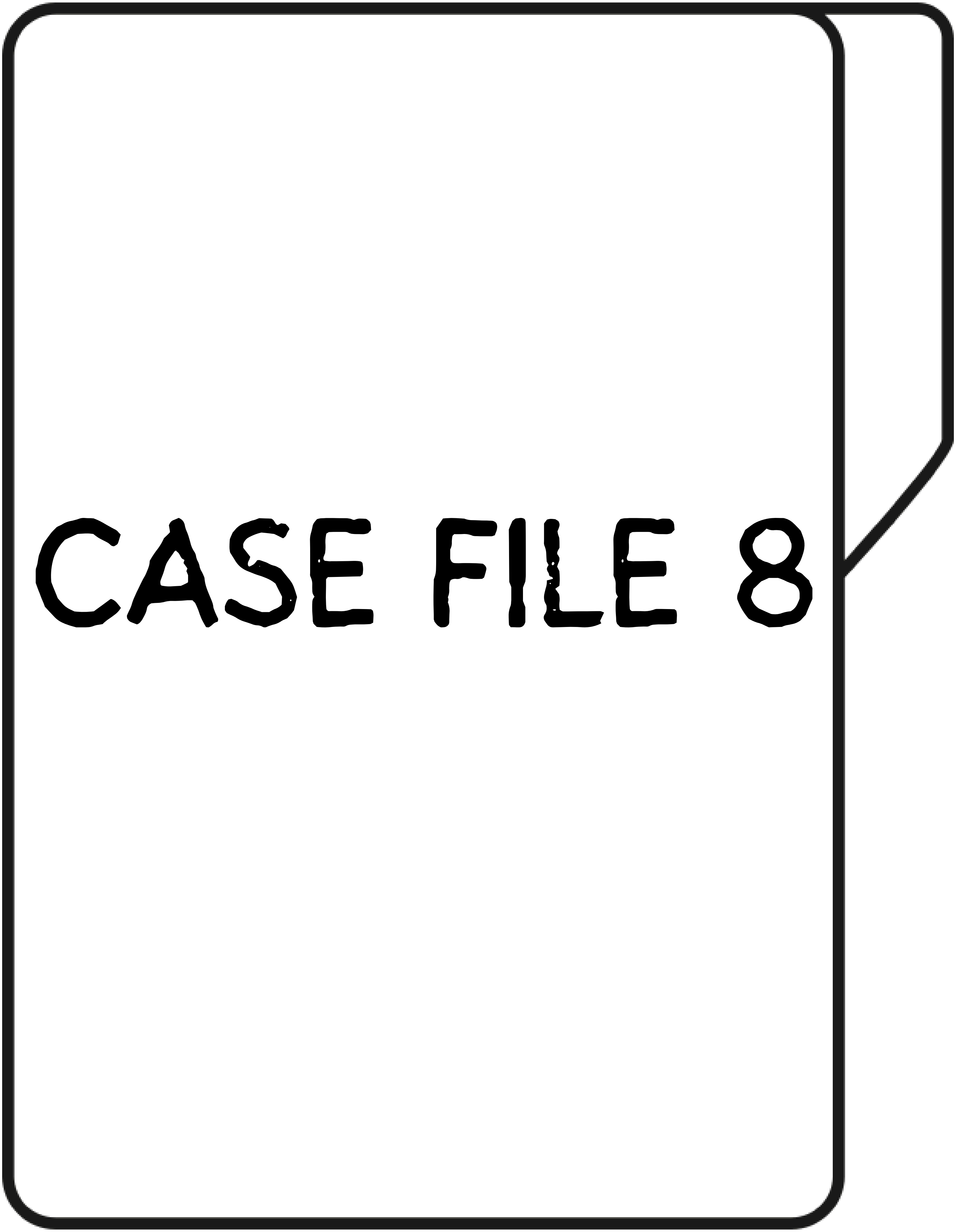

KĪLAUEA VOLCANO
LOWER EAST RIFT ZONE
LAVA FLOW HAZARD ZONES
(UPDATED: MAY 2024)
Pāhoa
130
11
Leilani Estates
Kapoho
Nanawale Estates
Vacationland Hawaii
EAST RIFT ZONE
Kīlauea Caldera
Pohoiki
Kalapana
11
Opihikao
PACIFIC OCEAN
HAWAI'I ISLAND
Area of Detail
LAVA FLOW HAZARD ZONES
Very High Likelihood
High Likelihood
Moderate Likelihood
Low Likelihood
MAP SYMBOLS
Community / Point of Interest
Major Road
Trail / Road (unpaved)
Stream / River
Coastline
Elevation Contours (Interval: 200 feet)
SCALE
0 2.5 5 7.5 10 km
0 1.5 3 4.5 6 mi
N
NOTES
Hazard zones show the relative likelihood that a lava flow from Kilauea Volcano could reach a particular area based on past eruptions, topography, and computer modeling. Hazard zones may change as new data become available.
SOURCES
Map data: U.S. Geological Survey (USGS)
Hawai'i Volcano Observatory (HVO)
Base map: USGS Topo & Imagery
Date: May 2024

Exhibit D

Updated map showing lava flow hazard zones, nearby communities, roads, and other geographic features around Kīlauea Volcano.



CASE FILE 8

CASE FILE 8

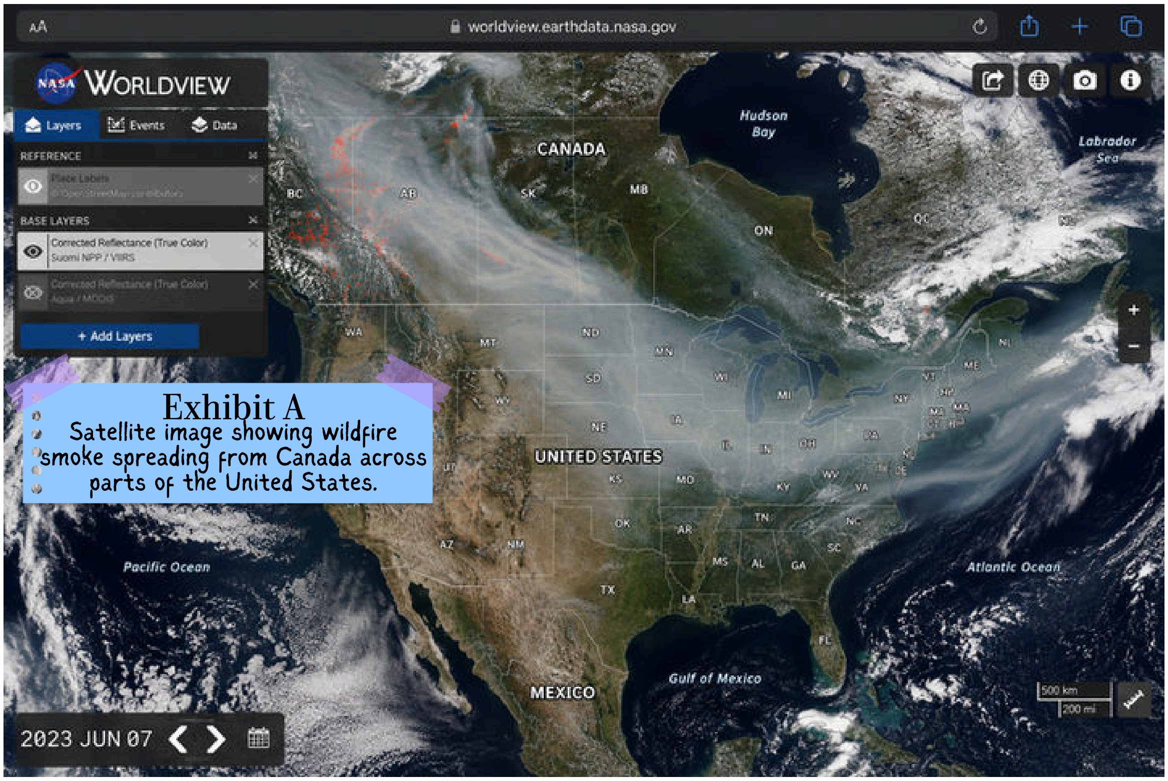
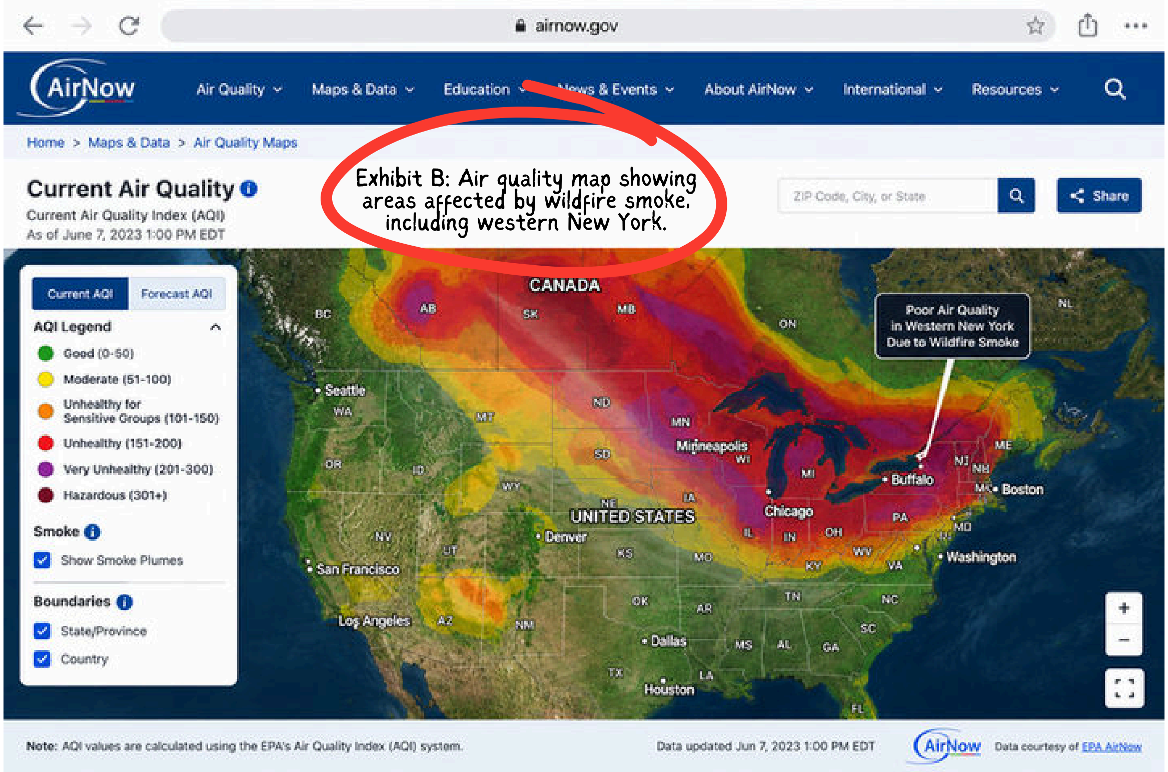

worldview.earthdata.nasa.gov
NASA WORLDVIEW
Layers
Events
Data
REFERENCE
Place Labels
BASE LAYERS
Corrected Reflectance (True Color)
Suomi NPP / VIIRS
Corrected Reflectance (True Color)
Aqua / MODIS
+ Add Layers
CANADA
Hudson Bay
Labrador Sea
BC
AB
SK
MB
ON
QC
NL
WA
MT
ND
MN
SD
WY
WI
MI
NE
IA
UNITED STATES
KS
MO
IL
IN
OH
PA
NY
VT
NH
ME
MA
KY
WV
VA
TN
NC
SC
OK
AR
MS
AL
GA
AZ
NM
TX
LA
FL
Pacific Ocean
Atlantic Ocean
Gulf of Mexico
MEXICO
500 km
200 mi
2023 JUN 07

Exhibit A
Satellite image showing wildfire smoke spreading from Canada across parts of the United States.


airnow.gov
AirNow
Air Quality
Maps & Data
Education
News & Events
About AirNow
International
Resources
Home > Maps & Data > Air Quality Maps
ZIP Code, City, or State
Share
Current AQI
Forecast AQI
CANADA
BC
AB
SK
MB
ON
NL
Seattle
WA
MT
ND
MN
Minneapolis
WI
MI
OR
ID
SD
WY
NE
IA
Chicago
Buffalo
NY
NH
ME
Boston
MA
PA
NJ
MD
Washington
OH
IN
IL
WV
VA
KY
UNITED STATES
Denver
NV
UT
San Francisco
Los Angeles
AZ
NM
KS
MO
OK
AR
TN
NC
SC
MS
AL
GA
Dallas
TX
LA
Houston
FL

Current Air Quality
Current Air Quality Index (AQI)
As of June 7, 2023 1:00 PM EDT

AQI Legend
- 🟩 Good (0-50)
- 🟨 Moderate (51-100)
- 🟧 Unhealthy for Sensitive Groups (101-150)
- 🟥 Unhealthy (151-200)
- 🟪 Very Unhealthy (201-300)
- 🟫 Hazardous (301+)

Smoke
☑ Show Smoke Plumes

Boundaries
☑ State/Province
☑ Country

Poor Air Quality in Western New York Due to Wildfire Smoke

Note: AQI values are calculated using the EPA's Air Quality Index (AQI) system.

Data updated Jun 7, 2023 1:00 PM EDT

Data courtesy of EPA AirNow

Exhibit B: Air quality map showing areas affected by wildfire smoke, including western New York.

CASE FILE 8

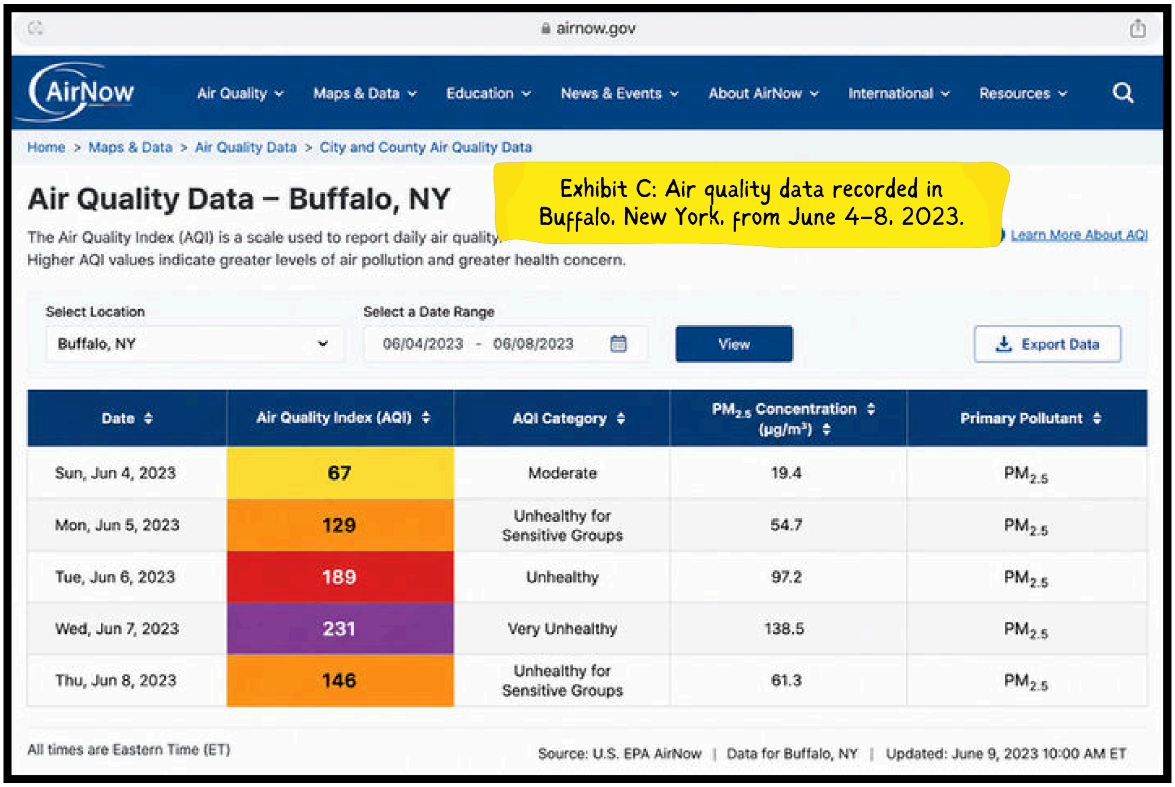


Exhibit C: Air quality data recorded in Buffalo, New York, from June 4–8, 2023.

airnow.gov
AirNow
Air Quality
Maps & Data
Education
News & Events
About AirNow
International
Resources
Home > Maps & Data > Air Quality Data > City and County Air Quality Data

Air Quality Data – Buffalo, NY

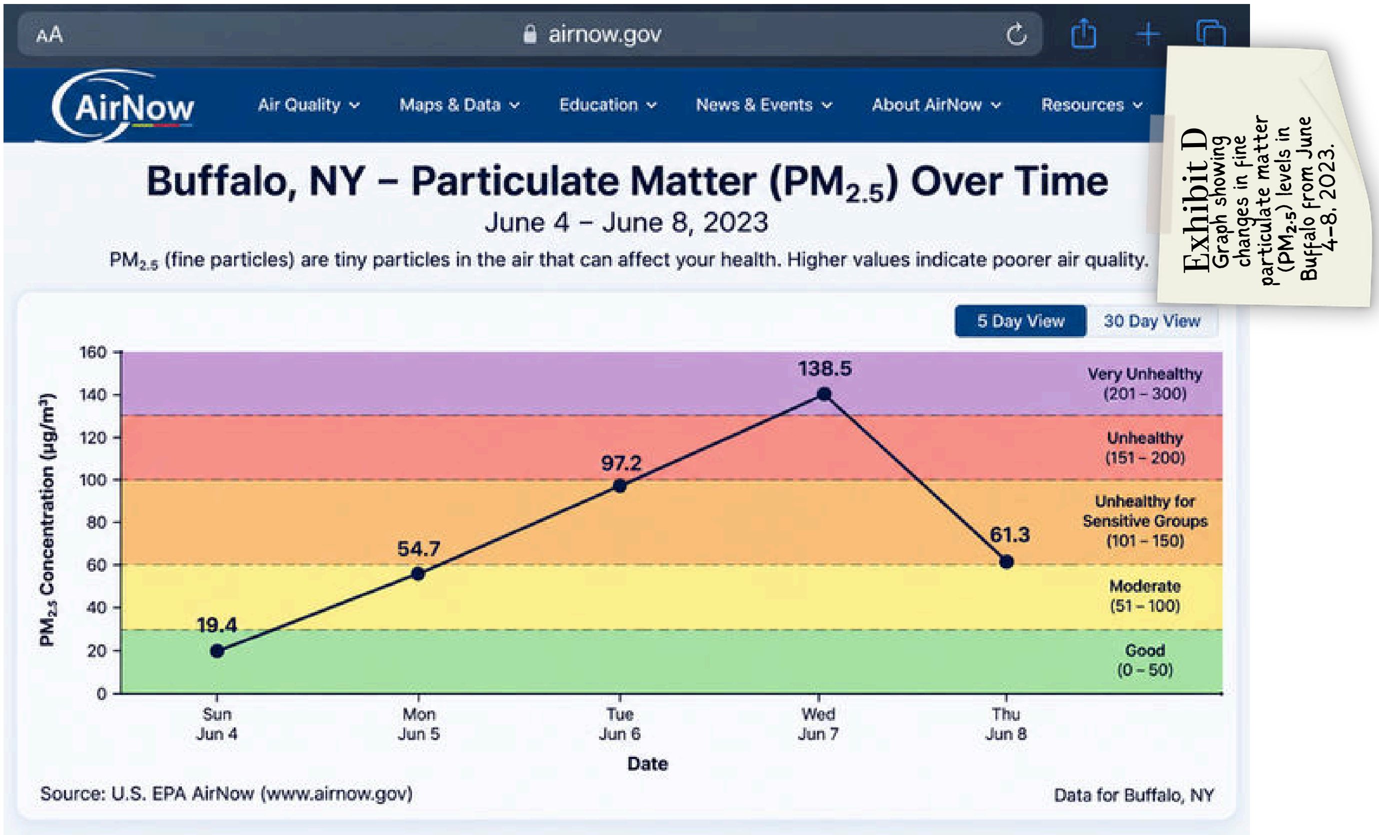
The Air Quality Index (AQI) is a scale used to report daily air quality. Higher AQI values indicate greater levels of air pollution and greater health concern.

Learn More About AQI
Select Location
Buffalo, NY
Select a Date Range
06/04/2023 - 06/08/2023
View
Export Data

Date	Air Quality Index (AQI)	AQI Category	$PM_{2.5}$ Concentration (µg/m³)	Primary Pollutant
Sun, Jun 4, 2023	67	Moderate	19.4	$PM_{2.5}$
Mon, Jun 5, 2023	129	Unhealthy for Sensitive Groups	54.7	$PM_{2.5}$
Tue, Jun 6, 2023	189	Unhealthy	97.2	$PM_{2.5}$
Wed, Jun 7, 2023	231	Very Unhealthy	138.5	$PM_{2.5}$
Thu, Jun 8, 2023	146	Unhealthy for Sensitive Groups	61.3	$PM_{2.5}$

All times are Eastern Time (ET)

Source: U.S. EPA AirNow | Data for Buffalo, NY | Updated: June 9, 2023 10:00 AM ET



Buffalo, NY – Particulate Matter ($PM_{2.5}$) Over Time

June 4 – June 8, 2023

$PM_{2.5}$ (fine particles) are tiny particles in the air that can affect your health. Higher values indicate poorer air quality.

Source: U.S. EPA AirNow (www.airnow.gov)

Data for Buffalo, NY

Exhibit D
Graph showing changes in fine particulate matter ($PM_{2.5}$) levels in Buffalo from June 4–8, 2023.